

MAT 1013: Discrete Mathematics

Session 11 : Relations

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Topics

- Cartesian products
- Binary relations
- Representations of relations
- Domain, codomain, image, and range
- Operations on relations
- Properties of relations
- Closures of relations
- Equivalence relations
- Partitions
- Partial orders
- Hasse diagrams
- Special elements in partially ordered sets
- Bounds in partially ordered sets
- Proof and problem-solving techniques

Learning Outcomes

By the end of this lesson, students should be able to:

- Explain the concept of relations and functions using correct mathematical notation;
- Define and use binary relations;
- Construct relations from Cartesian products;
- Represent relations using different representations;
- Determine the domain, codomain and range of a relation;
- Identify important properties of relations;
- Analyze the properties of relations, including reflexivity, symmetry, antisymmetry and transitivity.
- Construct and interpret equivalence relations and partial orders.
- Apply relations and functions in mathematical modelling.

1 Motivation

Before studying the formal definition of a relation, let us think about relationships that occur in everyday life.

Almost everything around us is connected to something else.

- Students are enrolled in courses.
- People have friends.
- Countries have capitals.
- Roads connect cities.
- Parents have children.
- Books have authors.
- Computers communicate through networks.

These are all examples of *relationships*.

The goal of this lesson is to understand these relationships using mathematics.

Key Question

How can we describe relationships between objects in a precise mathematical way?

Activity 1 : Relationships Around Us

For each of the following, identify

1. what objects are involved,
2. whether one object may be related to many others,
3. whether the relationship appears to be reversible.

Person	Friend
Student	Course
Country	Capital
Author	Book
Parent	Child
City	Road
Employee	Manager

One-to-One, One-to-Many and Many-to-Many Relationships

Not every relationship behaves in the same way.

- Some relationships associate one object with exactly one object.
- Others associate one object with many objects.
- Some relationships go both ways.

Example 1

A country has exactly one capital. Here are some examples

Sri Lanka $\longrightarrow$ Colombo

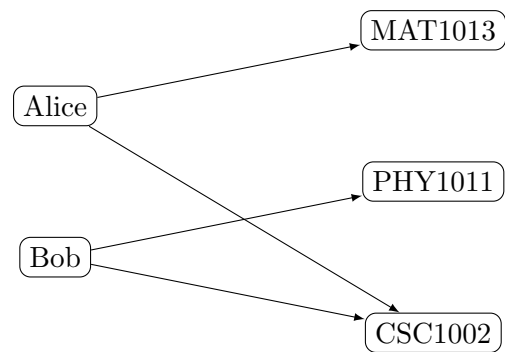
Japan $\longrightarrow$ Tokyo

France $\longrightarrow$ Paris

This looks very different from friendships and student - course relationship.

Example 2

One student may take many courses



Example 3

Friendship is usually mutual.

If Alice is a friend of Bob, then Bob is a friend of Alice.

Not every relationship has this property.

Different kinds of relationships possess different mathematical properties.
Our goal is to study these properties systematically.

Relations are built from Cartesian products.

Review of Cartesian Products

Let A and B be two sets. The Cartesian product of A and B is

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

The elements of a Cartesian product are **ordered pairs**.

Example 4

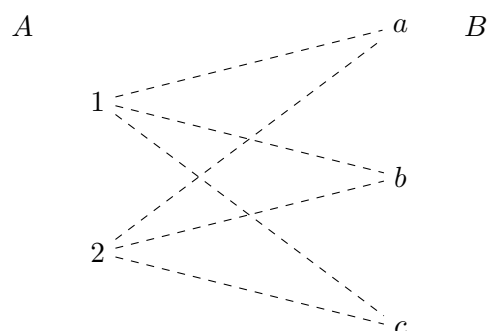
Let $A = \{1, 2\}$ and $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

Visualizing Cartesian Products

A Cartesian product may be viewed as every possible pairing between the two sets.

Every dashed line corresponds to one ordered pair.



Activity 2

Construct the following Cartesian products and visualize.

1. $\{0, 1\} \times \{x, y\}$
2. $\{a, b, c\} \times \{1, 2\}$
3. $\{1, 2, 3\} \times \{A, B\}$

Activity 3

Suppose $|A| = 3$ and $|B| = 4$.

Answer the following questions and discuss your observations before continuing.

1. How many ordered pairs belong to $A \times B$?
2. How many subsets does $A \times B$ have?
3. What might these subsets represent?

Looking Ahead

In the next section we shall discover that

every relation is simply a subset of a Cartesian product.

This simple idea forms the foundation of the entire chapter.

2 Binary Relations

In the previous section we saw that a Cartesian product contains *every possible ordered pair* between two sets.

In practice, however, we are usually interested in only some of these ordered pairs.

For example, let $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$.

The Cartesian product is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c)\}.$$

Suppose we are interested only in $\{(1, a), (2, b), (3, c)\}$.

This selected collection of ordered pairs is called a **relation**.

Definition: Binary Relation

Let A and B be two sets.

- A **binary relation** from A to B is any subset of the Cartesian product $A \times B$.

We write $R \subseteq A \times B$. or as $R : A \rightarrow B$

Reading the Notation

- A relation $R \subseteq A \times B$ is denoted by

$$R : A \rightarrow B.$$

- If $(a, b) \in R$, we say

a is related to b or aRb .

Both notations are common.

Example 5

Suppose $R = \{(1, 2), (2, 3), (3, 3)\}$. Then,

- $1R2$, meaning $(1, 2) \in R$
- It is not the case that $2R1$.
I.e., $(2, 1) \notin R$.

Example 7 : Divisibility

Let $A = \{1, 2, 3, 4, 6, 12\}$.

Define

$$aRb \iff a \mid b.$$

Then

$$(2, 4) \in R \text{ and } (3, 12) \in R, \\ \text{but } (4, 6) \notin R.$$

Example 9 : Friendship

Suppose $P = \{Alice, Bob, Carol, David\}$

Define

$$aRb \text{ whenever "a is a friend of b"}$$

Unlike divisibility, friendship is usually symmetric.

Can you draw the relation determined by the following descriptions:

Activity 4

For each of the following, identify the two sets involved and describe the relation.

1. Student enrolled in a course.
2. Country has a capital.
3. Person owns a vehicle.
4. Employee works in a department.
5. Book written by an author.
6. City connected by a highway.
7. Integer divides another integer.
8. A complex number has an absolute value.

Example 6: Less Than

Let $A = \{1, 2, 3, 4\}$.

Define

$$R = \{(a, b) : a < b\}.$$

Then,

$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

Example 8 : Congruence Modulo Four

Consider the integers. Define

$$aRb \iff a \equiv b \pmod{4}.$$

Examples include

$$2R6, \quad 5R1, \quad 10R2.$$

Later we shall prove that this is an *equivalence* relation.

How Many Relations Exist?

Suppose $|A| = m$ and $|B| = n$. Then

$$|A \times B| = mn.$$

Every relation is simply a subset of $A \times B$. Therefore,

$$\boxed{\text{Number of relations} = 2^{mn}.$$

Example 10

Suppose $|A| = 2$ and $|B| = 3$. Then $|A \times B| = 6$.

Hence, there are $2^6 = 64$ different relations from A to B .

Relations on a Set

If $A = B$ then the relation is called a **relation on the set A** .

Definition: Relation on a Set

A **relation on a set A** is a relation from A to A

Example 11

The following are all relations on a set.

“ $<$ ” : Less than

“ $|$ ” : divides (divisibility)

“ $\leq$ ” : Less than or equal

“ $\equiv \pmod{n}$ ” : Congruence modulo n

“ $=$ ” : is equal to (equality)

Visualizing Relations

There are several ways to represent exactly the same relation.

Throughout this lesson we shall use

1. Ordered pairs
2. Arrow diagrams
3. Tables
4. Matrices
5. Directed “graphs”

Each representation contains exactly the same mathematical information.

A relation is *not* the way we draw it. A relation is simply a set of ordered pairs. Arrow diagrams, tables, matrices, and graphs are merely different ways of displaying the same information.

Activity 5

Let $A = \{1, 2, 3\}$. Define, $R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}$.

In the next section we shall represent this relation

- as an arrow diagram,
- as a matrix,
- as a table,
- and as a directed graph.

Can you try to represent the above relation in a suitable way, so that one can visualize it.

Algorithmic Thinking

Algorithm : Constructing a Relation

Given two sets A and B . construct a relation as follows.

1. Construct the Cartesian product $A \times B$.
2. Decide which ordered pairs satisfy the desired property.
3. Select exactly those ordered pairs.
4. The selected subset is the required relation.

Section Summary

In this section we discovered that

- every relation is a subset of a Cartesian product,
- relations describe connections between objects,
- many different real-world situations can be modeled by relations,
- there are $2^{|A||B|}$ possible relations between two finite sets.

In the next section we shall learn how to represent a relation using five different mathematical representations.

3 Representing Relations

A relation is a collection (set) of ordered pairs. However, ordered pairs are not the only way to display a relation.

The same relation may be represented using:

1. a predicate or rule,
2. a set of ordered pairs,
3. an arrow diagram,
4. a table,
5. a relation matrix,
6. a directed graph.

These are not different relations. They are different representations of the same relation.

Representation 1: Predicate or Rule Form

Some relations are best described by a rule.

Examples 12

1. $R = \{(a, b) \in \mathbb{N}^2 : a < b\}$ defines the relation “less than” on $\mathbb{N}$.

2. Another example is

$$R = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod{3}\}.$$

This defines congruence modulo 3.

3. The relation

$$R = \{(a, b) : a \mid b\}$$

is such that a is related to b if a divides b .

4. All points on the unit circle is given by the following relation:

$$R = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

A predicate representation describes a relation by giving a condition that the ordered pairs must satisfy.

Activity 6

Write each relation in predicate form.

1. a is greater than or equal b .

3. a and b have same remainder modulo 5.

2. a is a factor of b .

4. x is the square of y .

Representation 2: Ordered Pairs

The following relation is given by listing all the ordered pairs of it.

$$R = \{(1, 2), (1, 4), (2, 2), (3, 1), (4, 3)\}.$$

This is the most direct representation because it explicitly lists every ordered pair in the relation.

Advantages

- Precise.
- Sometimes, it is easy to work with.
- Clearly shows which ordered pairs belong to the relation.

Disadvantages

- Becomes long for large relations.
- Not very visual.

Activity 7

Answer the following based on the above relation R .

1. Is $(1, 2) \in R$?
2. Is $(2, 1) \in R$?
3. Is $4R3$?
4. How many ordered pairs are in R ?

Representation 3: Arrow Diagram

An arrow diagram represents each ordered pair $(a, b) \in R$ by an arrow from a to b .

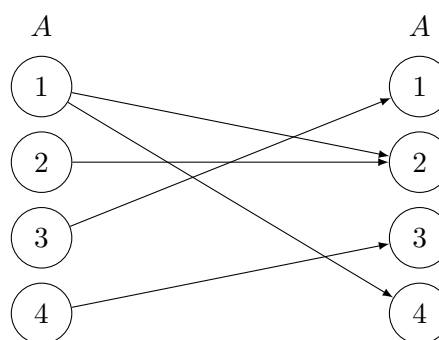
Example 13

The above relation R can be represented as follows.

Interpretation: The arrow from 1 to 2 represents $(1, 2) \in R$.

The arrow from 1 to 4 represents $(1, 4) \in R$

Likewise, each arrow represents an ordered pair in R .



Activity 8

Construct the arrow diagram for the relation $S = \{(1, 1), (1, 3), (2, 4), (4, 2)\}$ defined on the set $A = \{1, 2, 3, 4\}$.

Representation 4: Relation Table

A relation may also be displayed in a table.

- Rows represent the first coordinate.
- Columns represent the second coordinate.
- A check mark means the ordered pair belongs to the relation.

Example 14

R	1	2	3	4
1		✓		✓
2		✓		
3	✓			
4			✓	

The check mark in row 3, column 1 means

$$(3, 1) \in R.$$

Database Interpretation

A table representation is closely related to relational databases. Each row-column entry records whether a certain relationship exists.

Representation 5: Relation Matrix

The relation table may be converted into a matrix by replacing $\checkmark$ with 1, and blank entries with 0.

Example 15

The matrix of R is

$$M_R = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Interpretation :

The entry in row i , column j is 1 exactly when

$$(i, j) \in R.$$

For example, the entry in row 4, column 3 is 1, so

$$(4, 3) \in R.$$

Activity 9

Can you interpret $M_R^2 = M_R \cdot M_R$?

If you can't answer this, think of a very simple relation, convert it to the matrix form and then answer the same question in the context of your example.

George Pólya said:

“If you can't solve a problem, then there is an easier problem you can solve: find it.”

Algorithm: Constructing the Matrix of a Relation

Given a relation R on $A = \{a_1, a_2, \dots, a_n\}$:

1. Label the rows and columns by the elements of A .
2. Put 1 in position (i, j) if $(a_i, a_j) \in R$.
3. Put 0 otherwise.

Activity 10

Construct the relation matrix for $S = \{(1, 1), (1, 3), (2, 4), (4, 2)\}$ on $A = \{1, 2, 3, 4\}$.

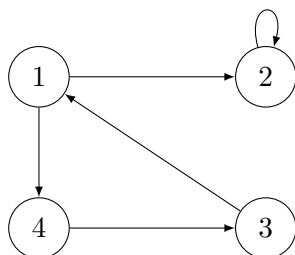
Representation 6: Directed Graph

A relation on a set may also be represented as a “directed graph”.

- Each element of the set becomes a vertex.
- Each ordered pair $(a, b) \in R$ becomes a directed edge from a to b .

Example 16

Above relation R on $\{1, 2, 3, 4\}$ can be represented by a “directed graph” as follows:



Observations :

From the directed graph, we can see:

- there is a loop at 2;
- there is an arrow from 1 to 2;
- there is an arrow from 3 to 1;
- there is no arrow from 2 to 1.

- Here, by a “Graph” we mean a collection of vertices and some edges between these vertices.
- This may look different from the graph of a function, even though both have a mathematical similarity. Only the representation is different.
- Directed graphs will later become one of the main objects of study in graph theory.

Converting Between Representations

The six representations are equivalent. We should be able to move from one representation to another.

Example 17 : From Ordered Pairs to Matrix

Given $R = \{(1, 2), (1, 4), (2, 2), (3, 1), (4, 3)\}$,

place 1 in exactly those positions:

$$M_R = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Exaple 18 : From Matrix to Ordered Pairs

Suppose

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}.$$

The 1's occur in positions: $(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)$.

Thus,

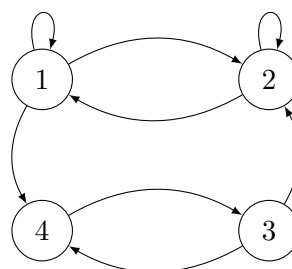
$$R = \{(1, 1), (1, 3), (2, 2), (3, 1), (3, 2)\}.$$

Assume the underlying set is $A = \{1, 2, 3\}$.

Activity 11 : From Digraph to Ordered Pairs

To recover ordered pairs from a directed graph:

1. Look at every arrow.
2. If there is an arrow from a to b , write (a, b) .
3. Collect all such ordered pairs.



Recover the ordered pairs of the relation given by the following directed graph.

Comparison of Representations

Representation	Advantage	Limitation
Predicate	Best for infinite relations	May be abstract
Ordered pairs	Precise and direct	Long for large sets
Arrow diagram	Easy to understand visually	Messy for large sets
Table	Organized for finite relations	Not suitable for large/infinite sets
Matrix	Useful for computation	Needs ordering of elements
Directed graph	Excellent for graph theory	Can become cluttered

Historical and Computational Note

Relations in Computer Science

Relations are central in computer science.
Relational databases store data using tables.
Graph databases store data using nodes and edges.
Relation matrices are used in algorithms, networks, data science, and graph theory.
Thus, the six representations studied here are not merely theoretical. They are different ways of organizing and computing with relational information.

Common Mistakes

Common Mistakes

1. Thinking that different diagrams represent different relations.
2. Forgetting that ordered pairs are ordered:
$$(a, b) \neq (b, a) \text{ in general.}$$
3. Confusing a relation with a function.
4. Confusing the matrix entry m_{ij} with the ordered pair (i, j) or (j, i) .

Activity 12

Let $A = \{1, 2, 3, 4\}$ and $S = \{(1, 1), (1, 3), (2, 4), (3, 2), (4, 4)\}$.

Complete the following:

1. Write S as a table.
2. Write the relation matrix M_S .
3. Draw an arrow diagram.
4. Draw a directed graph.
5. Describe S in words, if possible.

Section Summary

Summary

A relation can be represented in many ways:

1. *Predicate*
2. *Ordered Pairs*
3. *Arrow Diagram*
4. *Table*
5. *Matrix*
6. *Directed Graph*

All these forms contain exactly the same information.

4 Domain, Codomain, Image and Range

In the previous section, we saw that a relation can be represented in many different ways: by ordered pairs, rules, arrow diagrams, tables, matrices, and directed graphs.

We now study four important pieces of information associated with a relation:

domain, codomain, image, range.

These ideas are especially important when we later study functions.

Motivation

Consider the following relation from students to courses.

$$A = \{\text{Alice, Bob, Carol, David}\}$$

$$B = \{\text{Math, Physics, Computing, Statistics}\}.$$

Suppose the relation $R \subseteq A \times B$ is given by

$$R = \{(\text{Alice, Math}), (\text{Alice, Computing}), (\text{Bob, Physics}), (\text{Carol, Computing})\}.$$

This relation tells us which student is enrolled in which course.

Notice the following:

- David appears in the set of students, but David is not related to any course.
- Statistics appears in the set of courses, but no student is related to Statistics.
- Alice is related to two courses.
- Computing is related to two students.

This shows that we need to distinguish carefully between the set of possible inputs, the set of possible outputs, and the set of outputs that are actually used.

Key Question

When a relation is given from A to B , which elements of A are actually used, and which elements of B are actually reached?

Domain and Codomain

Definition: Domain and Codomain

Let $R \subseteq A \times B$ be a relation from A to B .

- The set A is called the **domain set** or **source set**.
- The set B is called the **codomain** or **target set**.

Sometimes, authors use the word **domain** to mean the set of all first coordinates that actually appear in the relation. To avoid confusion, in this handout we distinguish:

“source set” A from “actual domain” of R .

Actual Domain of a Relation

Definition: Actual Domain

Let $R \subseteq A \times B$ be a relation from A to B .

The **actual domain** (or the **active domain**) of R , denoted $\text{Dom}(R)$, is

$$\text{Dom}(R) = \{a \in A : \text{there exists } b \in B \text{ such that } (a, b) \in R\}.$$

In words, $\text{Dom}(R)$ is the set of elements of A that are actually related to at least one element of B .

Thus,

$$\text{Dom}(R) \subseteq A.$$

The actual domain may be smaller than the source set.

Range of a Relation

Definition: Range

Let $R \subseteq A \times B$ be a relation.

The **range** (or the **image**) of R , denoted $\text{Ran}(R)$ (or $\text{Img}(R)$), is

$$\text{Ran}(R) = \{b \in B : \text{there exists } a \in A \text{ such that } (a, b) \in R\} = \text{Img}(R).$$

In words, $\text{Ran}(R)$ is the set of elements of B that are actually reached by the relation.

Thus,

$$\text{Ran}(R) \subseteq B.$$

The range may be smaller than the codomain.

Image of an Element

Definition: Image of an Element

If $R \subseteq A \times B$ and $a \in A$, the **image of a** under R (denoted by $R(a)$) is the set of all elements of B related to a .

I.e., For $a \in A$,

$$R(a) = \{b \in B : (a, b) \in R\}.$$

For a general relation, $R(a)$ may contain:

- no elements,
- one element,
- many elements.

These become an important difference between general “relations” and the special class of relations called the “functions”.

Image of a Subset

Definition: Image of a Subset

If $S \subseteq A$, the image of S under R is

$$R(S) = \{b \in B : \text{there exists } a \in S \text{ such that } (a, b) \in R\}.$$

Equivalently,

$$R(S) = \bigcup_{a \in S} R(a).$$

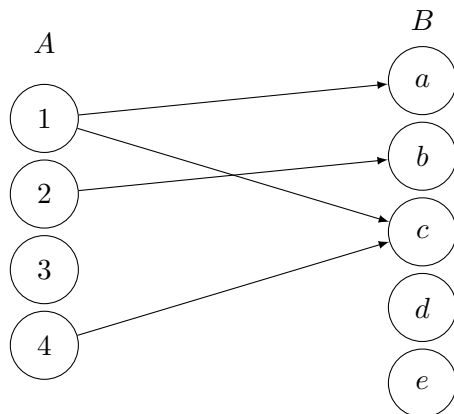
In particular,

$$R(A) = \text{Ran}(R).$$

Example 19: Visual Interpretation

Consider the following relation from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c, d, e\}$.

$$R = \{(1, a), (1, c), (2, b), (4, c)\}.$$



From the diagram:

$$\text{Dom}(R) = \{1, 2, 4\}.$$

The element 3 is not in the actual domain because it has no outgoing arrow.

Also,

$$\text{Ran}(R) = \{a, b, c\}.$$

The elements d and e are in the codomain, but not in the range.

Furthermore,

$$\begin{aligned} R(1) &= \{a, b\}, & R(2) &= \{b\}, \\ R(3) &= \emptyset, & R(4) &= \{c\}. \end{aligned}$$

Pre-image of an Element

The “preimage” look backwards in the relation. I.e., for a given element $b \in B$, the preimage of b is the set of all elements in A that relates to b .

Definition: Pre-image of an Element

If $R \subseteq A \times B$ and $b \in B$, the **preimage of b** under R (denoted by $R^{-1}(b)$) is the set of all elements of A related to b .

I.e., For $b \in B$,

$$R^{-1}(b) = \{a \in A : (a, b) \in R\}.$$

Pre-image of a Subset

Definition: Image of a Subset

If $T \subseteq B$, the preimage of T under R is

$$R^{-1}(T) = \{a \in A : \text{there exists } b \in T \text{ such that } (a, b) \in R\}.$$

Equivalently,

$$R^{-1}(T) = \bigcup_{b \in T} R^{-1}(b).$$

In particular,

$$R^{-1}(B) = \text{Dom}(R).$$

Example 20

In the above Example 19,

$$R^{-1}(c) = \{1, 4\} \quad \text{and} \quad R^{-1}(d) = \emptyset.$$

Also,

$$R^{-1}(\{a, c\}) = \{1, 4\}, \quad R^{-1}(\{b, c, d\}) = \{1, 2, 4\} \quad \text{and} \quad R^{-1}(\{d, c\}) = \emptyset.$$

Example 21

Let $A = \{1, 2, 3, 4\}$, $B = \{x, y, z\}$ and $R \subseteq A \times B$ be $R = \{(1, x), (1, y), (2, z), (4, x)\}$.

Determine the following.

- | | | |
|--------------------|-----------|-----------------------|
| 1. $\text{Dom}(R)$ | 3. $R(1)$ | 5. $R^{-1}(x)$ |
| 2. $\text{Ran}(R)$ | 4. $R(3)$ | 6. $R^{-1}(\{x, y\})$ |

Solution : The first coordinates appearing in R are 1, 2, 4. Therefore, $\text{Dom}(R) = \{1, 2, 4\}$.

The second coordinates appearing in R are x, y, z . Therefore, $\text{Ran}(R) = \{x, y, z\}$.

Since $(1, x), (1, y) \in R$ we have $R(1) = \{x, y\}$.

Since there is no ordered pair in R whose first coordinate is 3, $R(3) = \emptyset$.

Because the elements of A that relates to $x \in B$ are 1 and 4, we have that $R^{-1}(x) = \{1, 4\}$.

We also have that $R^{-1}(\{x, y\}) = \{1, 4\}$.

Example 22

Let $A = \{1, 2, 3, 4, 6, 12\}$. Define a relation R on A by: $aRb \iff a \mid b$.

Find $R(2)$ and $R^{-1}(12)$.

Solution : Since R is a relation on A , only elements of A are considered.

Now $R(2) = \{b \in A : 2 \mid b\}$. The elements of A divisible by 2 are 2, 4, 6, 12. Therefore, $R(2) = \{2, 4, 6, 12\}$.

Also, $R^{-1}(12) = \{a \in A : a \mid 12\}$. The elements of A that divides 12 are 1, 2, 3, 4, 6, 12. Therefore, $R^{-1}(12) = \{1, 2, 3, 4, 6, 12\} = A$.

Remark : $5 \notin A$. So $R(5)$ is not defined in this relation, because 5 is not an element of the source set.

Example 23

Let $R = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$. This relation is defined from $\mathbb{R}$ to $\mathbb{R}$.

The actual domain is $\text{Dom}(R) = \mathbb{R}$, because every real number x has a square.

The range is $\text{Ran}(R) = [0, \infty)$, because $x^2 \geq 0$ for every real number x , and every nonnegative real number occurs as a square.

For example, $R(2) = \{4\}$ and $R(-2) = \{4\}$.

For every positive real number y (i.e., $y > 0$) there are two real numbers x such that $(x, y) \in R$. Namely, $\pm\sqrt{y}$. Thus, for $y > 0$, $R^{-1}(y) = \{\pm\sqrt{y}\}$.

Also, $R^{-1}(0) = \{0\}$ and for $y < 0$, $R^{-1}(y) = \emptyset$.

Example 24: A Relation That Is Not a Function

Let $R = \{(1, a), (1, b), (2, b), (3, c)\}$.

Then $R(1) = \{a, b\}$. Thus, the element 1 is related to two different elements. This is allowed for a relation.

Later, when we define functions, we shall see that a function does not allow one input to have two different outputs.

Example 25: Domain and Range from Relation Matrices

Suppose R is a relation from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c, d\}$, and the relation matrix is

$$M_R = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Rows correspond to elements of A , and columns correspond to elements of B .

- **Finding the Actual Domain:**

- An element of A belongs to $\text{Dom}(R)$ if its row contains at least one 1.

Rows 1, 3, and 4 contain at least one 1. Row 2 contains only zeros.

Therefore,

$$\text{Dom}(R) = \{1, 3, 4\}.$$

- **Finding the Range:**

- An element of B belongs to $\text{Ran}(R)$ if its column contains at least one 1.

Columns a , b , and c contain at least one 1. Column d contains only zeros.

Therefore,

$$\text{Ran}(R) = \{a, b, c\}.$$

Algorithmic Thinking: Finding Domain and Range

Algorithmic Thinking: Domain and Range from Ordered Pairs

Given a relation $R \subseteq A \times B$, written as a set of ordered pairs:

1. Collect all first coordinates appearing in the ordered pairs. This gives $\text{Dom}(R)$.
2. Collect all second coordinates appearing in the ordered pairs. This gives $\text{Ran}(R)$.
3. The codomain is not necessarily the same as the range. It must be given as part of the relation from A to B .

Algorithmic Thinking: Domain and Range from a Relation Matrix

Given a relation matrix:

1. A row contributes to the actual domain if it contains at least one 1.
2. A column contributes to the range if it contains at least one 1.
3. Rows with only zeros correspond to source elements not used by the relation.
4. Columns with only zeros correspond to codomain elements not reached by the relation.

Activity 13

Can you give similar algorithms to find the domain, range, image and pre-image for different types of representations of a relation?

Common Mistakes

Common Mistakes

Mistake 1: Confusing Codomain and Range

- The codomain and range are not necessarily equal.
- The codomain is the set of all possible target values.
- The range is the set of target values actually reached.

Mistake 2: Confusing Domain and Range

- The domain consists of first coordinates.
- The range consists of second coordinates.
- Do not interchange them.

Mistake 3: Assuming Every Element Must Be Used

- For $R \subseteq A \times B$, not every element of A need to appear as a first coordinate.
(This will change when we later discuss functions from A to B , where every input in A must have exactly one output.)
- Also, not every element of B must appear as a second coordinate.

Example 26

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $R = \{(1, a), (2, b), (3, a)\}$.

1. The codomain is $B = \{a, b, c, d\}$. But the range is $\text{Ran}(R) = \{a, b\}$. Thus, $\text{Ran}(R) \neq B$.
2. We have $\text{Dom}(R) = \{1, 2, 3\}$, while $\text{Ran}(R) = \{a, b\}$.
3. Note that $\text{Dom}(R)$ is a proper subset of A , and $\text{Ran}(R)$ is also a proper subset of B .

Comparison Table

Let $R \subseteq A \times B$.

Concept	Meaning
Source set A	The set from which first coordinates are allowed.
Codomain B	The set from which second coordinates are allowed.
Actual domain $\text{Dom}(R)$	Elements of A that actually occur as first coordinates.
Range $\text{Ran}(R)$	Elements of B that occur as second coordinates.
Image $R(a)$	All elements in B related to a particular $a \in A$.
Image $R(S)$	All elements reached from elements of $S \subseteq A$.
Pre-image $R^{-1}(b)$	All elements in A related to $b \in B$.
Pre-image $R^{-1}(T)$	All elements in A related some $b \in T$.

Exercises

1. Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $R = \{(1, a), (1, b), (2, b), (3, c), (4, a)\}$. Determine the following.

(a) $R(1)$ (b) $R(4)$ (c) $R(\{1, 4\})$ (d) $R^{-1}(\{b, d\})$

2. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{x, y, z, w\}$ and $R = \{(1, x), (1, z), (2, y), (3, y), (5, w)\}$.

Find:

(a) $\text{Dom}(R)$ (c) $R(1)$ (e) $R(\{1, 2\})$
 (b) $\text{Ran}(R)$ (d) $R^{-1}(y)$ (f) $R^{-1}(\{y, w\})$

3. Let $R = \{(x, y) \in \mathbb{R}^2 : y = x + 1\}$. Determine the following.

(a) $\text{Dom}(R)$ (b) $\text{Ran}(R)$ (c) $R^{-1}([0, \infty))$ (d) $R([0, \infty))$

4. $R = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$. Determine the following:

(a) $\text{Dom}(R)$ (b) $\text{Ran}(R)$ (c) $R([0, \infty))$ (d) $R^{-1}((-1, 1))$

5. Let $R = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}$. Determine the following:

(a) $\text{Dom}(R)$ (b) $\text{Ran}(R)$ (c) $R([0, \infty))$ (d) $R^{-1}((-1, 1))$

6. Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $R = \{(1, a), (1, b), (3, c)\}$. Find

(a) the actual domain (c) range (e) $R(2)$
 (b) codomain (d) $R(1)$ (f) $R^{-1}(\{c, d\})$

7. Let $R = \{(x, y) \in \mathbb{Z}^2 : x < y\}$. Find:

(a) $R(2)$

(b) $R^{-1}(2)$

(c) Is the actual domain all of $\mathbb{Z}$?

(d) Is the range all of $\mathbb{Z}$?

8. Let $R = \{(x, y) \in \mathbb{R}^2 : y = \sqrt{x}\}$. Find the actual domain and range.

9. Let $A = \{1, 2, 3, 4, 5, 6\}$ and define aRb iff $a \mid b$. Find $R(2)$, $R^{-1}(3)$, and $R^{-1}(\{2, 3\})$.

10. Given the relation matrix

$$M_R = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

on $A = \{1, 2, 3\}$, find the actual domain and range.

11. Let $R = \{(x, y) \in \mathbb{R}^2 : y = x^2 - 4\}$. Find the actual domain and range.

12. Let $R = \{(x, y) \in \mathbb{R}^2 : x = y^2 - 4\}$. Find the actual domain and range.

13. Give examples:

(a) Give an example of a relation $R \subseteq A \times B$ such that $\text{Dom}(R) = A$ but $\text{Ran}(R) \neq B$.

(b) Give an example of a relation $R \subseteq A \times B$ such that $\text{Dom}(R) \neq A$ but $\text{Ran}(R) = B$.

(c) Give an example of a relation where some element $a \in A$ satisfies $|R(a)| = 3$.

Section Summary

Summary

For a relation $R \subseteq A \times B$,

- A is the source set.
- B is the codomain.
- $\text{Dom}(R)$ is the set of first coordinates actually used.
- $\text{Ran}(R)$ is the set of second coordinates actually reached.
- $R(a)$ is the set of all elements related to a .
- $R(S)$ is the set of all elements reached from a subset $S \subseteq A$.

Do not confuse codomain with range. $\text{Ran}(R) \subseteq B$, but equality need not hold.

Looking Ahead

Now that we know how to identify the domain and range of a relation, we are ready to study operations on relations.

In the next part we shall define the inverse R^{-1} of a relation R , and the composition $S \circ R$ of two relations R and S .

These operations allow us to reverse and combine relationships.

5 Operations on Relations

In the previous sections we learned how to describe a relation, represent it in different ways, and identify its domain, codomain and range.

We now study how to *operate* on relations.

The main operations are:

- inverse relation,
- composition of relations,
- identity relation,
- union, intersection and difference of relations.

These operations allow us to reverse, combine and compare relationships.

Motivation

Suppose we have the relation R consisting of pairs of the form (Student, Course). For example, this relation may mean:

student is enrolled in course.

The inverse relation would mean:

course has student enrolled.

So reversing a relation reverses the direction of the relationship.

Now suppose we also have another relation S consisting of pairs of the form (Course, Lecturer); meaning

course is taught by lecture.

Then by combining the two relations, we can form a new relation:

student is taught by lecturer.

This is an example of composition of relations.

Inverse Relation

Given a relation $R \subseteq A \times B$, the inverse relation of R , denoted R^{-1} , is the relation from B to A given by reversing all the arrows of R .

In particular,

$$(a, b) \in R \iff (b, a) \in R^{-1}.$$

Definition: Inverse Relation

The inverse of a relation $R \subseteq A \times B$ is the relation $R^{-1} \subseteq B \times A$ given by:

$$R^{-1} = \{(b, a) \in B \times A : (a, b) \in R\}.$$

Example 27

Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ and $R = \{(1, a), (1, c), (2, b), (3, a)\}$.

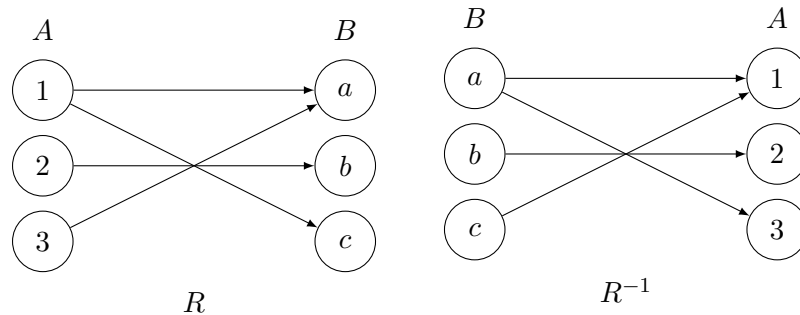
Then

$$R^{-1} = \{(a, 1), (c, 1), (b, 2), (a, 3)\}.$$

Notice that $R \subseteq A \times B$, while $R^{-1} \subseteq B \times A$.

Example 28 : Arrow Diagram Interpretation

The inverse relation reverses every arrow.

**Example 29 : Matrix Interpretation**

If R is a relation from $A = \{a_1, \dots, a_m\}$ to $B = \{b_1, \dots, b_n\}$, then the matrix of R is an $m \times n$ matrix.

The matrix of R^{-1} is an $n \times m$ matrix.

In fact, the matrix of R^{-1} is the transpose matrix.

$$M_{R^{-1}} = M_R^T,$$

(The transpose of a matrix M , denoted by M^T , is obtained by writing i -th row as the i -th column.)

Example 30:

$$\text{If } M_R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \text{ then } M_{R^{-1}} = M_R^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Theorem

Theorem: Inverse of the inverse

For every relation R ,

$$(R^{-1})^{-1} = R.$$

Proof. We need to show the two sets R and $(R^{-1})^{-1}$ are equal. We shall show that one is contained in the other.

Let $(a, b) \in (R^{-1})^{-1}$.

Conversely, let $(a, b) \in R$.

By the definition of inverse relation,

Then $(b, a) \in R^{-1}$.

$$(b, a) \in R^{-1}.$$

Therefore, $(a, b) \in (R^{-1})^{-1}$.

Again, by the definition of inverse relation

Hence $R \subseteq (R^{-1})^{-1}$.

$$(a, b) \in R.$$

. Thus $(R^{-1})^{-1} \subseteq R$.

Therefore, $(R^{-1})^{-1} = R$. □

Alternative proof

We can give another proof based on our matrix view point. First note that for a matrix M , $(M^T)^T = M$.

Proof: Method 2. We have that

$$M_{(R^{-1})^{-1}} = M_{R^{-1}}^T = (M_R^T)^T = M_R.$$

Therefore, $(R^{-1})^{-1} = R$. □

Activity 14

Let $A = \{1, 2, 3, 4\}$, $B = \{x, y, z\}$, and $R = \{(1, x), (1, z), (3, y), (4, x)\}$.

Find:

1. R^{-1}

3. $\text{Ran}(R^{-1})$

2. $\text{Dom}(R^{-1})$

4. the matrix of R^{-1}

Composition of Relations

Recall our example where R relates students to courses, and S relates courses to lecturers. Then we can combine them to obtain a relation that relates students to lecturers. This combined relation is called the **composition**.

Definition: Composition of Relations

Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The composition of R followed by S , denoted by

$$S \circ R,$$

is the relation from A to C defined by

$$S \circ R = \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Thus,

$(a, c) \in S \circ R$ if there exists an intermediate element $b \in B$ such that aRb and bSc .

Common Mistake

In $S \circ R$, the relation R is applied first, and then S .
This is the same order convention used for composition of functions.

Composability

Note that two relations R and S are composable in the order $S \circ R$ if and only if the codomain of R is contained in the domain of S .

Example 31

Let $A = \{1, 2, 3\}$, $B = \{a, b\}$ and $C = \{x, y, z\}$.

Let $R = \{(1, a), (2, a), (2, b), (3, b)\}$ and $S = \{(a, x), (a, y), (b, z)\}$.

Note that, as per the definition, we can compose R and S in the order $S \circ R$.

Find $S \circ R$.

Solution :

We look for paths of the form $A \longrightarrow B \longrightarrow C$.

From $(1, a) \in R$ and $(a, x), (a, y) \in S$, we obtain $(1, x), (1, y) \in S \circ R$.

From $(2, a) \in R$ and $(a, x), (a, y) \in S$, we obtain $(2, x), (2, y) \in S \circ R$.

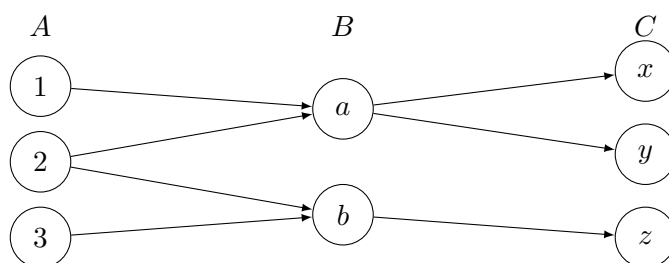
From $(2, b) \in R$ and $(b, z) \in S$, we obtain $(2, z) \in S \circ R$.

From $(3, b) \in R$ and $(b, z) \in S$, we obtain $(3, z) \in S \circ R$.

Therefore,

$$S \circ R = \{(1, x), (1, y), (2, x), (2, y), (2, z), (3, z)\}.$$

Arrow Diagram Interpretation



Each two-step path from A to C gives an ordered pair in $S \circ R$.

Algorithmic Thinking: Algorithm for Composition

Algorithm: Computing $S \circ R$

Given $R \subseteq A \times B$ and $S \subseteq B \times C$

To compute $S \circ R$:

1. Start with an ordered pair $(a, b) \in R$.
2. Find all ordered pairs $(b, c) \in S$ with the same middle element b .
3. For each such pair, include (a, c) in $S \circ R$.
4. Repeat for every pair in R .

Composition and Matrices

For finite relations, composition corresponds to Boolean matrix multiplication.

If M_R is the matrix of R , and M_S is the matrix of S , then the matrix of $S \circ R$ is obtained by Boolean multiplication:

$$M_{S \circ R} = M_R \odot M_S.$$

Here, ordinary addition is replaced by logical “or”, and ordinary multiplication is replaced by logical “and”.

This idea becomes important in graph theory and computer science.

Example 32 : Composition via Boolean Matrix Product

$$\text{Let } M_R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } M_S = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then the entry in row 2, column 3 of $M_{S \circ R}$ is obtained by checking whether there is a middle element connecting 2 to 3.

Row 2 of M_R is $(1, 1)$ and column 3 of M_S is $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Since $(1 \wedge 0) \vee (1 \wedge 1) = 1$, the entry is 1.

$$\text{The full Boolean product is } M_{S \circ R} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Activity 15

Let $A = \{1, 2\}$, $B = \{a, b, c\}$ and $C = \{x, y\}$.

Let $R = \{(1, a), (1, b), (2, c)\}$ and $S = \{(a, x), (b, y), (c, x), (c, y)\}$.

1. Draw an arrow diagram for the composition $S \circ R$.
2. Hence determine $S \circ R$.
3. Determine the matrix of $S \circ R$.
4. Writedown the matrix of R and S and verify that the boolean product yields the same matrix as before.

Identity Relation and Its Properties

The identity relation I_A on a set A relates every element to itself and to no other element.

$$aI_Ab \iff a = b.$$

Definition: Identity Relation

Let A be a set. The identity relation on A , denoted I_A , is

$$I_A = \{(a, a) : a \in A\}.$$

Example 33

If $A = \{1, 2, 3\}$, then $I_A = \{(1, 1), (2, 2), (3, 3)\}$

The matrix of I_A is $M_{I_A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. This is in fact the identity matrix.

Identity Relation and Composition

The identity relation behaves like the number 1 in multiplication.

If $R \subseteq A \times B$, then $R \circ I_A = R$ and $I_B \circ R = R$.

Composing with the identity relation does not change a relation.

Theorem

Theorem

If $R \subseteq A \times B$, then $R \circ I_A = R$ and $I_B \circ R = R$

Proof. We will prove $R \circ I_A = R$. The other case, $I_B \circ R = R$ can be proved in a similar way.

Let $(a, b) \in R \circ I_A$

Then there exists $x \in A$ such that

$$(a, x) \in I_A \text{ and } (x, b) \in R.$$

Since $(a, x) \in I_A$, we have $a = x$.

Therefore, $(a, b) = (x, b) \in R$.

Hence $R \circ I_A \subseteq R$

Conversely, let $(a, b) \in R$.

Note that $(a, a) \in I_A$.

We have that there exists $a \in A$ such that

$$(a, a) \in I_A \text{ and } (a, b) \in R.$$

Therefore, $(a, b) \in R \circ I_A$

Thus, $R \subseteq R \circ I_A$

Hence, $R \circ I_A = R$

□

Activity: 16

Can you prove the above result using the matrix representation of a relation and the boolean matrix multiplication for composition.

Union, Intersection and Difference of Relations

Since relations are sets of ordered pairs, we may apply ordinary set operations to them.

Let $R, S \subseteq A \times B$. Then

$$R \cup S, \quad R \cap S, \quad R \setminus S$$

are also relations from A to B .

Example 34

Let $R = \{(1, a), (2, b), (3, c)\}$ and $S = \{(1, a), (1, b), (3, c), (4, a)\}$. Then,

$$R \cup S = \{(1, a), (2, b), (3, c), (1, b), (4, a)\},$$

$$R \cap S = \{(1, a), (3, c)\},$$

$$R \setminus S = \{(2, b)\}.$$

Activity 17

Let $R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}$ and $S = \{(1, 2), (2, 2), (3, 1), (3, 3)\}$.

Find:

1. $R \cup S$

3. $R \setminus S$

5. R^{-1}

2. $R \cap S$

4. $S \setminus R$

6. S^{-1}

6 Useful Results About Relation Operations

***This section is mainly for extra reading**

Proof Strategies for Relations

Relations are sets of ordered pairs. Therefore, many proof techniques from set theory apply directly to relations.

Proof Strategy: Proving Equality of Relations

Proof Strategy: Equality of Relations

To prove that two relations R and S are equal, prove both inclusions:

$$R \subseteq S \quad \text{and} \quad S \subseteq R.$$

Since relations are sets of ordered pairs, this means:

- Take an arbitrary ordered pair $(a, b) \in R$ and show that $(a, b) \in S$.
- Then take an arbitrary ordered pair $(a, b) \in S$ and show that $(a, b) \in R$.

Thinking Like a Mathematician

Thinking Like a Mathematician

Diagrams are useful to think about relations.

However, to prove equality of relations, we get insight from diagrams but may not use them directly.

Instead, we compare membership of ordered pairs.

This is exactly the same method used to prove equality of sets.

Theorem

Theorem

For relations R and S ,

$$(R \cup S)^{-1} = R^{-1} \cup S^{-1}.$$

Proof. We will prove the two inclusions to show the equality of sets.

Let $(a, b) \in (R \cup S)^{-1}$.

Then $(b, a) \in R \cup S$.

Therefore, $(b, a) \in R$ or $(b, a) \in S$.

Hence, $(a, b) \in R^{-1}$ or $(a, b) \in S^{-1}$.

Thus, $(a, b) \in R^{-1} \cup S^{-1}$.

So $(R \cup S)^{-1} \subseteq R^{-1} \cup S^{-1}$.

Conversely, let $(a, b) \in R^{-1} \cup S^{-1}$.

Then, $(a, b) \in R^{-1}$ or $(a, b) \in S^{-1}$.

Hence, $(b, a) \in R$ or $(b, a) \in S$.

So $(b, a) \in R \cup S$.

Therefore, $(a, b) \in (R \cup S)^{-1}$.

So $R^{-1} \cup S^{-1} \subseteq (R \cup S)^{-1}$.

Therefore, $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

□

Theorem

Theorem

For composable relations R and S ,

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$

Proof. Suppose $R \subseteq A \times B$ and $S \subseteq B \times C$.

Let $(a, c) \in A \times C$

$$\begin{aligned}(a, c) \in (S \circ R)^{-1} &\iff (c, a) \in S \circ R \\ &\iff \text{exists } b \text{ such that } (c, b) \in R \text{ and } (b, a) \in S \\ &\iff \text{exists } b \text{ such that } (b, c) \in R^{-1} \text{ and } (a, b) \in S^{-1} \\ &\iff (a, c) \in R^{-1} \circ S^{-1}.\end{aligned}$$

Thus, $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$. □

Theorem : Associativity of Composition

Theorem

If R , S , and T are composable relations, then

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Remark

This means that when composing three relations, the placement of parentheses does not matter, as long as the order of the relations is preserved.

This is similar to associativity of function composition.

Common Mistakes

Common Mistakes

1. Confusing R^{-1} with the reciprocal of a number.
2. Forgetting that inverse relations reverse ordered pairs.
3. Writing $R \circ S$ when the intended order is $S \circ R$.
4. Forgetting that in $S \circ R$, R is applied first.
5. Assuming relation composition is commutative. In general,

$$S \circ R \neq R \circ S.$$

Exercises (ctd.)

14. Let $R = \{(1, a), (2, b), (3, a)\}$. Find R^{-1} .
15. Let $R = \{(1, 2), (2, 3), (3, 4)\}$ on $A = \{1, 2, 3, 4\}$. Find R^{-1} and draw both directed graphs.
16. Let $R = \{(1, a), (2, a), (2, b)\}$ and $S = \{(a, x), (b, y)\}$. Find $S \circ R$.
17. Let $R = \{(1, 2), (2, 3)\}$ and $S = \{(2, 1), (3, 2)\}$. Compute $R \cup S$, $R \cap S$, and $R \setminus S$.
18. Let $A = \{1, 2, 3\}$. Write down the identity relation I_A .
19. Verify directly that $(R^{-1})^{-1} = R$ for $R = \{(1, a), (2, b), (3, a)\}$.
20. Some challenges.
 - (a) Give an example where $S \circ R \neq R \circ S$.
 - (b) Prove that $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.
 - (c) Prove that $(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}$.
 - (d) Let $R \subseteq A \times B$. Prove that $\text{Dom}(R^{-1}) = \text{Ran}(R)$.
 - (e) Let $R \subseteq A \times B$. Prove that $\text{Ran}(R^{-1}) = \text{Dom}(R)$.
 - (f) Prove associativity of relation composition: $T \circ (S \circ R) = (T \circ S) \circ R$.

Section Summary

Summary

In this section we studied operations on relations.

- The inverse relation R^{-1} reverses every ordered pair.
- The composition $S \circ R$ combines two relations through an intermediate set.
- The identity relation I_A relates every element of A to itself.
- Relations can be combined using union, intersection and difference.
- Relation operations satisfy useful algebraic rules.

Important identities include:

$$(R^{-1})^{-1} = R, \quad (R \cup S)^{-1} = R^{-1} \cup S^{-1}, \quad (S \circ R)^{-1} = R^{-1} \circ S^{-1}, \\ R \circ I_A = R, \quad I_B \circ R = R.$$

Looking Ahead

This completes the foundational language of relations. We now know how to:

construct relations, represent relations, identify domain and range,

reverse relations, compose relations.

The next natural question is: “What special properties can a relation possess?”

- Can every object be related to itself?
- Can every arrow be reversed?
- Can relationships be chained?

These questions lead to:

reflexive, symmetric, antisymmetric, and transitive relations.

and eventually to

equivalence relations, partial orders and functions.

7 Reflexive Relations

Motivation

Suppose the relation is

“is the same age as.”

Is every person the same age as themselves? Certainly yes.

Now consider the relation

“is equal to.”

Every number is equal to itself.

Similarly,

every set is a subset of itself.

These examples suggest an important property.

Key Question

Given a relation, is every object related to itself?

Definition

A “Reflexive Relation” on a set is a relation where every element is related to itself.

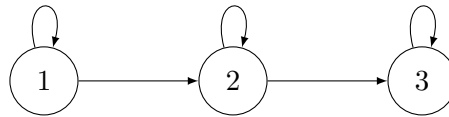
Definition: Reflexive Relation

A relation R on a set A is called **reflexive** if

$$(a, a) \in R \text{ for every } a \in A.$$

Visualization

A reflexive relation always contains a loop at every vertex, when represented as a directed graph.



Notice that every vertex has a loop. So, the above relation represented as a directed graph is a reflexive relation

Matrix Interpretation

Theorem

A relation matrix represents a reflexive relation if and only if every diagonal entry is equal to 1.

Example 35

Suppose $A = \{1, 2, 3\}$. The matrix $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ represents a reflexive relation because every diagonal entry is 1.

Example 36 : A reflexive relation

Let $A = \{1, 2, 3\}$ and
 $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$.

Since $(1, 1), (2, 2), (3, 3) \in R$, the relation is reflexive

Example 37: A non reflexive relation

Consider $R = \{(1, 1), (2, 2), (1, 2)\}$.
 Since $(3, 3) \notin R$, the relation is not reflexive.

Algorithmic Thinking

Algorithm: Testing Reflexivity

Given a relation R on a finite set A ,

1. For each $a \in A$, check whether $(a, a) \in R$
2. If you find a pair $(a, a) \notin R$, then you have found a counter example and R is not reflexive.
 Otherwise, every diagonal pair appears, and the relation is reflexive.

Activity 18

Determine whether each of the following relations are reflexive.

1. $R = \{(a, a), (b, b), (c, c), (d, d), (a, b), (c, d)\}$ on $A = \{a, b, c, d\}$.
2. $\{(1, 1), (2, 2), (3, 3)\}$ on $\{1, 2, 3\}$.
3. $\{(1, 1), (2, 2)\}$ on $\{1, 2, 3\}$.
4. Divisibility relation on $\{1, 2, 3, 6\}$.

5. “Less than” relation ($<$) on $\{1, 2, 3\}$.

6. “Less than or equal” relation ($\leq$) on $\mathbb{Z}$.

Activity 19

Construct the smallest reflexive relation on $A = \{1, 2, 3, 4, 5\}$.

How many ordered pairs does it contain?

Example 38

The following relations often considered in mathematics are reflexive.

Subset relation	$\subseteq$	Equality	$=$
Divisibility	$a \mid b$	Less than or equal to	$\leq$
Congruence modulo n	$a \equiv b \pmod{n}$	Greater than or equal to	$\geq$

Section Summary

Summary

- A relation is reflexive if $(a, a) \in R$ for every element of the set.
- Think of reflexivity as *Every object is related to itself*.
- Graphically, every vertex has a loop.
- In the relation matrix, every diagonal entry is equal to 1.

8 Irreflexive Relations

Motivation

Consider the relation “*is less than*.”

Certainly, for any number x , it is never the case that $x < x$.

Likewise, no person is older than themselves.

These examples motivate another important property of relations.

Definition

A relation on a set is said to be *irreflexive* (or *anti-reflexive*) if no element is related to itself. I.e., it does not contain any of the elements of the form (a, a) .

Definition: Irreflexive /Anti-reflexive Relations

A relation R on a set A is called **irreflexive** (or **anti-reflexive**) if

$$(a, a) \notin R \text{ for every } a \in A.$$

Example 39

Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (1, 3), (2, 3)\}$.

Since $(1, 1), (2, 2), (3, 3) \notin R$, the relation is irreflexive.

Example 40

Consider $R = \{(1, 2), (2, 2)\}$.

Since $(2, 2) \in R$ the relation is not irreflexive.

Note that $(1, 1) \notin R$, so it is also not reflexive.

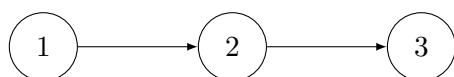
So, “irreflexive” does not mean “not reflexive”.

Visualization

An irreflexive relation has *no loops at any vertex*.

Example 41

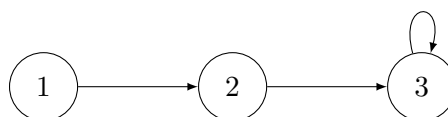
The relation given by the following directed graph is irreflexive.



It does not contain any loop.

Example 42

The relation given by the following directed graph is neither reflexive nor irreflexive.



It does not contain all loops, so it is not reflexive.

It contains a loop, so it is not irreflexive.

Matrix Interpretation

A relation matrix represents an irreflexive relation precisely when every diagonal entry is 0.

Example 43

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

has zeros on the diagonal and therefore represents an irreflexive relation.

Example 44

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \text{ has 1's on the diagonal,}$$

so it is not irreflexive.

Also, not all diagonal entries are 1. Therefore, it is also not reflexive.

Example 45

There are relations that are neither reflexive nor irreflexive. For example, consider the relation $R = \{(1, 1), (1, 2)\}$ on $A = \{1, 2, 3\}$.

The relation is not reflexive because $(2, 2)$ and $(3, 3)$ are missing.

The relation is not irreflexive because $(1, 1)$ is present.

Hence it is neither reflexive nor irreflexive.

Algorithmic Thinking

Algorithm: Testing Irreflexivity

For every element $a \in A$, check whether (a, a) belongs to the relation.
If even one diagonal pair is present, then the relation is not irreflexive.
Otherwise, it is irreflexive.

Activity 20

Construct the largest irreflexive relation on $A = \{1, 2, 3, 4\}$.

How many ordered pairs does it contain?

Activity 21

Determine whether each of the following relation is irreflexive.

1. $R = \{(a, b), (b, c), (c, d)\}$ on $\{a, b, c, d\}$
2. $<$ on $\{1, 2, 3, 4\}$
3. $>$ on $\mathbb{Z}$
4. Equality
5. Divisibility
6. Congruence modulo 5

Comparing Reflexive and Irreflexive Relations

The two concepts are opposites in a precise sense, but they should not be confused with the idea of “not reflexive.”

Property	Reflexive	Irreflexive
Definition	$(a, a) \in R$ for every a	$(a, a) \notin R$ for every a
Loops in digraph	Every vertex has a loop	No vertex has a loop
Diagonal of matrix	All 1's	All 0's
Smallest possible relation	Identity relation	Empty relation
Typical examples	$=, \leq, \subseteq$	$<, >, \subset$

Given a relation, there are three possibilities:

1. Reflexive.
2. Irreflexive.
3. Neither reflexive nor irreflexive.

Do not confuse “not reflexive” with “irreflexive.”

Common Mistakes

Common Mistakes

Note that “not reflexive” **does not mean** “irreflexive.”
A relation can be neither reflexive nor irreflexive.

Example 45

The following relations often considered in mathematics are irreflexive.

Proper subset	$\subset$	Less than	$<$
Strict divisibility	$a \mid b$ and $a \neq b$	Greater than	$>$

Section Summary

Summary

An irreflexive relation satisfies $(a, a) \notin R$ for every element.
Graphically, there are no loops.
In the matrix, every diagonal entry is 0.

9 Symmetric Relations

Motivation

consider the relation “*is a friend of*.”

If Alice is a friend of Bob, then we usually expect Bob to be a friend of Alice.

Similarly, if two people are siblings, then each is a sibling of the other.

These relationships possess an important property called **symmetry**.

Definition

A relation on a set is *symmetric* when one object is related to another, the reverse relationship also holds. In other words, whenever one arrow exists the reverse arrow must also exist.

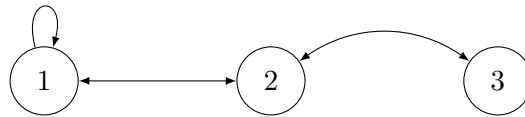
Definition: Symmetric Relation

A relation R on a set A is called **symmetric** if

$$(a, b) \in R \implies (b, a) \in R \quad \text{for every } a, b \in A$$

Visualization

A symmetric relation always contains arrows in pairs (i.e., both ways).



Note that every arrow has a matching arrow in the opposite direction. We have represented both of these arrows by a single bidirectional arrow.

Matrix Interpretation

A relation is symmetric precisely when its relation matrix is symmetric about the main diagonal.

Theorem

Theorem

A matrix M is symmetric iff $M^T = M$.

Example 46

Suppose $A = \{1, 2, 3\}$. The matrix $M = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ is symmetric because $m_{ij} = m_{ji}$ for every pair of indices.

This means when ever there is an arrow from i to j , there is also an arrow from j to i . Thus, if $(i, j) \in R$, then $(j, i) \in R$.

Thus, the corresponding relation is also symmetric.

Example 47

Let $A = \{1, 2, 3\}$ and
 $R = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$.

Every ordered pair has its reverse.

Therefore, the relation is symmetric.

Example 48

Consider $R = \{(1, 2), (2, 3)\}$.

Becuause $(1, 2) \in R$, but $(2, 1) \notin R$ the relation is not symmetric.

Algorithmic Thinking

Algorithm: Testing Symmetry

For every ordered pair $(a, b) \in R$, check whether $(b, a) \in R$.
 If every pair has its reverse, the relation is symmetric.
 Otherwise, it is not symmetric.

Activity 22

Consider the relation $R = \{(a, b), (b, a), (a, c), (c, a), (b, b)\}$.

1. Check whether the relation R is symmetric.
2. Can you improve your algorithm to reduce the number of steps required.
3. Is constructing the relation matrix makes it easier?
4. What can you say about the inverse relation?

Activity 23

Determine whether each relation is symmetric.

- | | |
|--|-------------------------|
| 1. Equality on the real numbers | 4. Congruence modulo 5 |
| 2. Less than ($<$) on the integers | 5. Divisibility |
| 3. Less than or equal ($\leq$) on the integers | 6. Parallelism of lines |

Activity 24

What is the smallest symmetric relation on $A = \{1, 2, 3, 4\}$?

What is the smallest symmetric relation on A that contain ordered pairs $(1, 2), (2, 4), (3, 1)$?

Example 49

The following relations often considered in mathematics are **symmetric**.

Equality	=
Congruence modulo n	
Parallel lines	

Example 50

The following relation often considered in mathematics are **not symmetric**.

Less than	$<$
Less than or equal to	$\leq$
Divisibility	$ $
Subset relation	$\subseteq$

Activity 25

Construct the largest symmetric relation on $A = \{1, 2, 3\}$ that is not reflexive.

Theorem

Theorem

A relation R is symmetric if and only if $R = R^{-1}$ if and only if $M_R^T = M_R$ (i.e., the matrix of the relation is symmetric)

Proof. We will prove R is symmetric iff $R = R^{-1}$. Then, the other equivalence follows from the properties of inverse relation.

We need to show two directions: “if” and “only if” parts.

($\implies$): Suppose R is symmetric.

For any ordered pair (a, b) , we have that.

$$(a, b) \in R \iff (b, a) \in R$$

($\because R$ is symmetric)

$$\iff (a, b) \in R^{-1}$$

(by the definition of R^{-1})

Hence, every element of R belongs to R^{-1} ,

and every element of R^{-1} belongs to R .

Therefore, $R = R^{-1}$

($\impliedby$): Suppose $R = R^{-1}$

Suppose $(a, b) \in R$

Then, $(b, a) \in R^{-1}$

(by the definition of R^{-1}).

Then, $(b, a) \in R$ (because $R = R^{-1}$).

Thus R is symmetric.

Thus, R is symmetric iff $R = R^{-1}$. □

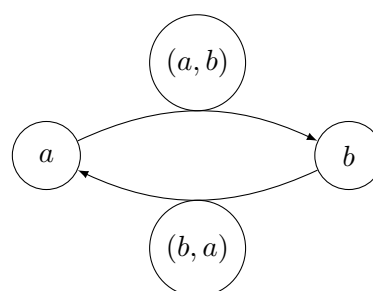
Visual Interpretation of the Theorem

If $(a, b) \in R$ then the directed graph of R contains an arrow $a \longrightarrow b$.

The inverse relation R^{-1} reverses every arrow: Thus, $b \longrightarrow a$ is in the graph of R^{-1} .

Therefore, if $R = R^{-1}$, then every arrow appears together with its reverse.

This is exactly symmetry.



Symmetry: every arrow has a reverse arrow

Comparing Reflexive, Irreflexive and Symmetric Relations

Property	Reflexive	Irreflexive	Symmetric
Loops	Every vertex	No vertex	Optional
Diagonal of matrix	All 1's	All 0's	No restriction
Reverse arrows	Not required	Not required	Required
Typical examples	$=, \leq$	$<$	$=$, Congruence

Common Mistakes

Common Mistakes

Students often check only one or few arrows. Symmetry requires **every** arrow to have its reverse.

One missing reverse arrow means the relation is not symmetric. (I.e., Every ordered pair must have its reverse.)

Section Summary

Summary

A relation is symmetric whenever $(a, b) \in R \iff (b, a) \in R$.

Graphically: Every arrow has a matching reverse arrow.

Matrix representation: $M_R = M_R^T$

Via Inverse Relations: $R = R^{-1}$

10 Antisymmetric Relations

Motivation

In the previous section we learned that a symmetric relation allows arrows to occur in pairs.

Now consider a different situation.

Consider the relation “*is less than or equal to*.”

Certainly, $2 \leq 5$ is true; however, $5 \leq 2$ is false.

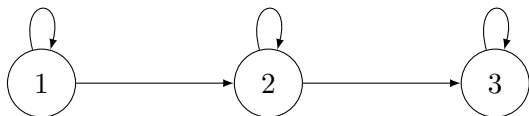
Likewise, for *divisibility* relation $2 \mid 6$ is true, but $6 \mid 2$ is false.

These examples suggest another important kind of relation.

Antisymmetric (Anti-Symmetric) Relations

Example 51

Consider the following digraph.



Note that, there are no arrows in opposite directions between different vertices.

This is the essential idea behind antisymmetry.

Definition

Definition: Antisymmetry

A relation R on a set A is called **antisymmetric** if

$$(a, b) \in R \text{ and } (b, a) \in R \text{ together imply } a = b.$$

Equivalent Ways to Think

If the formal definition looks intimidating, here are different interpretations of the definition that you might find easy to understand.

Equivalent ways of thinking about Antisymmetry:

- Whenever two distinct elements are related, they cannot point to each other.
I.e., If $a \neq b$, then $(a, b) \notin R$ or $(b, a) \notin R$.
- Two different vertices are never allowed to have arrows in both directions.
Loops are perfectly acceptable.

Visualization

Allowed:



Not allowed:



Matrix Interpretation

The matrix of an antisymmetric relation has the property that

$$\text{if } m_{ij} = 1 \text{ and } m_{ji} = 1, \text{ then } i = j.$$

Equivalently, considering the contrapositive,

if $i \neq j$, then both of the off-diagonal entries at (i, j) and (j, i) cannot be 1 at the same time.

Example 52

Let $A = \{1, 2, 3\}$ and
 $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$.

There are no distinct elements connected in both directions.

Hence, R is antisymmetric.

Example 53

Consider $R = \{(1, 2), (2, 1), (3, 2), (3, 3)\}$.

Note that $1 \neq 2$ and both $(1, 2)$ and $(2, 1)$ are in R .

Therefore, the relation is not antisymmetric.

Algorithmic Thinking

Algorithm: Testing Antisymmetry

For every pair of distinct elements $a \neq b$
check whether $(a, b) \in R$ and $(b, a) \in R$.
If such a pair exists, then relation is not antisymmetric.
Otherwise, it is antisymmetric.

Activity 26

Consider $R = \{(1, 2), (2, 3), (1, 3)\}$. Check whether R is antisymmetric.

Example 54

The following relations often considered in mathematics are **antisymmetric**.

Equality
Less than or equal
Greater than or equal
Subset
Divisibility
Proper divisibility

Example 55

The following relations often considered in mathematics are **not antisymmetric**.

Congruence modulo 5
Parallel lines

Example 56

Note that equality is both “symmetric” and “antisymmetric”.

In particular, “antisymmetric” **does not mean** “not symmetric”.

Activity 27

Determine whether each of the following relations is antisymmetric.

- | | | |
|---------------|------------------------|-------------------------|
| 1. Equality. | 3. Less than or equal. | 5. Congruence modulo 4. |
| 2. Less than. | 4. Divisibility. | 6. Subset relation |

Activity 28

Discuss how the following relations that we encounter in day-to-day life can be classified.

- | | | |
|---------------|----------------|---------------|
| 1. Sibling of | 2. Friendship. | 3. Parent of. |
|---------------|----------------|---------------|

Activity 29

Construct a largest antisymmetric relation on $A = \{1, 2, 3\}$.

How many ordered pairs does it contain?

Common Mistakes

Common Mistakes

Loops do not violate antisymmetry. Many students incorrectly believe that (a, a) is forbidden (which is false).

Antisymmetry only restricts *distinct* elements to be not connected both ways.

Theorem

Theorem

The subset relation $\subseteq$ on sets is antisymmetric.

Proof. To prove “ $\subseteq$ ” is antisymmetric, we need to show for any two sets A and B , if $A \subseteq B$ and $B \subseteq A$ then $A = B$. But, this is the definition of set equality.

In detail:

Suppose $A \subseteq B$ and $B \subseteq A$.

Then every element of A belongs to B , and every element of B belongs to A .

Therefore, the two sets contain exactly the same elements.

Hence $A = B$

Thus “ $\subseteq$ ” is antisymmetric. □

Symmetric versus Antisymmetric versus Not symmetric

One of the most common misconceptions is the belief that “Antisymmetric means not symmetric”. This is not correct as we see in the following activities and examples.

In fact, a relation may be

- symmetric but not antisymmetric,
- both symmetric and antisymmetric,
- antisymmetric but not symmetric,
- neither symmetric nor antisymmetric.

Let us carefully examine these differences.

Activity 30

(Do this before reading through.) Can you think of a relation that is

- symmetric?
- both?
- antisymmetric?
- neither?

Activity 31

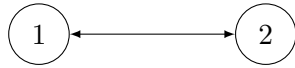
For each relation below, discuss whether it is

- symmetric
- both
- antisymmetric
- neither

- | | | |
|-----------------|----------------------------|---------------------|
| 1. Equality | 4. Congruence modulo 5 | 7. Parent of |
| 2. Less than | 5. Friendship | 8. Is a sibling of |
| 3. Divisibility | 6. “Has the same birthday” | 9. Is a creditor of |

Example 57: A Symmetric but not Antisymmetric

Consider $A = \{1, 2\}$ and $R = \{(1, 2), (2, 1)\}$.



The relation is symmetric because every arrow has its reverse.

It is not antisymmetric because there are arrows both ways between 1 and 2.

Example 58: An Antisymmetric but not Symmetric

Suppose $R = \{(1, 2)\}$.



Notice that there is no reverse arrow. Therefore, the relation is antisymmetric.

Also, it is not symmetric because there is no reverse arrow from 2 to 1.

Example 59: Both Symmetric and Antisymmetric

Consider $R = \{(1, 1), (2, 2)\}$.

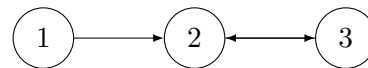


Every arrow is a loop and loops reverse to themselves. So this relation is symmetric.

There are no arrows joining different vertices. Therefore, this relation is also antisymmetric.

Example 60: Neither Symmetric nor Antisymmetric

Consider $R = \{(1, 2), (2, 3), (3, 2)\}$ on $A = \{1, 2, 3\}$.



The reverse arrows from 2 to 1 is missing, so it is not symmetric.

It is also not antisymmetric because there are arrows in both ways between 3 and 2.

Example 61: A Classification Table

Relation	Symmetric	Antisymmetric
Equality	✓	✓
Less than	×	✓
Less than or equal	×	✓

Relation	Symmetric	Antisymmetric
Congruence modulo n	✓	×
Subset	×	✓
Identity relation	✓	✓
Friendship	Usually ✓	×

Example 62 : A Classification Table (another way)

Here is another way to summarize them.

It also shows that symmetry and antisymmetry are independent properties.

	Symmetric	Not Symmetric
Antisymmetric	Equality	$\leq$, $<$, Subset, Divisibility
Not Antisymmetric	Congruence, Parallel Lines, Same Birthday	Parent Of, Teacher Of

Section Summary

Summary

Antisymmetry means both $(a, b) \in R$ and $(b, a) \in R$ can happen only when $a = b$.
Think of antisymmetry as

Two different objects cannot point to each other.

Loops are allowed.

The words *symmetric* and *antisymmetric* are misleading.

Antisymmetric does not mean “not symmetric.”

Neither one imply the other.

Symmetry and antisymmetry are independent properties.

A relation can be:

- symmetric but not antisymmetric,
- antisymmetric but not symmetric,
- both symmetric and antisymmetric,
- neither symmetric nor antisymmetric.

11 Concept Consolidation: Comparing Relation Properties

We have now encountered four important properties of relations:

- | | |
|---------------|-----------------|
| • Reflexive | • Symmetric |
| • Irreflexive | • Antisymmetric |

Although their definitions are short, one can easily confuse these properties. The following summary brings together the essential ideas.

Comparison Table

Property	Definition	Digraph Interpretation	Interpretation	Memory Aid
Reflexive	Every element satisfies $(a, a) \in R$	Every vertex has a loop.		Everyone is related to themselves.
Irreflexive	Every element satisfies $(a, a) \notin R$	No loops.		Nobody is related to themselves.
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	Every arrow has a reverse arrow.		What goes out comes back.
Antisymmetric	$(a, b), (b, a) \in R \Rightarrow a = b$	No two different vertices may point to each other.		Different objects cannot point both ways.

Matrix Interpretation

The four properties can also be recognized directly from the relation matrix.

Property	Matrix Interpretation
Reflexive	Every diagonal entry is 1.
Irreflexive	Every diagonal entry is 0.
Symmetric	The matrix is symmetric. Equivalently, the matrix equals its transpose: $M = M^T$
Antisymmetric	Whenever $m_{ij} = 1$ and $m_{ji} = 1$ then necessarily $i = j$. Equivalently, there are no symmetric off-diagonal pairs of 1's.

Graphical Summary



Algorithmic Checklist

Whenever you are given a relation, ask the following questions.

- | | |
|--|----------------|
| 1. Does every element have a loop? | Reflexive? |
| 2. Does every element have no loop? | Irreflexive? |
| 3. Does every arrow have a reverse arrow? | Symmetric? |
| 4. Are any two different vertices not joined in both directions? | Antisymmetric? |

Important Observations

Things to Remember

1. Reflexive and irreflexive are opposite concepts.
2. A relation may be neither reflexive nor irreflexive.
3. Symmetric and antisymmetric are *not* opposites.
4. A relation may be
 - symmetric but not antisymmetric,
 - antisymmetric but not symmetric,
 - both,
 - neither.
5. Equality is both symmetric and antisymmetric.

Activity 32

Classify each relation by completing the following table

Relation	Reflexive	Irreflexive	Symmetric	Antisymmetric
Equality				
Less than				
Less than or equal				
Congruence modulo 5				
Divisibility				
Friendship				
Subset				
Proper subset				
Parent of				

Exercises (Ctd.)

Answer the following without calculation. Use the definitions.

21. Can a relation be both reflexive and symmetric?
22. Can a relation be reflexive but not symmetric?
23. Can a relation be symmetric but not reflexive?
24. Can a relation be both reflexive and irreflexive, if the underlying set is nonempty?
25. Can a relation be neither reflexive nor irreflexive?
26. Can a relation be symmetric and antisymmetric at the same time?
27. Does symmetric mean “not antisymmetric”?
28. Does antisymmetric mean “not symmetric”?

Looking Ahead

One important property still remains.

Suppose aRb and bRc .

Should this imply aRc ?

This leads to one of the most important concepts in discrete mathematics:

Transitivity

Transitivity, together with reflexivity and symmetry, gives rise to equivalence relations.

Transitivity, together with reflexivity and antisymmetry, gives rise to partial orders.

12 Transitive Relations

Motivation

Many mathematical relationships possess an important feature known as *transitivity*.

For example,

if $5 < 8$ and $8 < 12$ then certainly $5 < 12$.

Similarly, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

The conclusion follows by combining two existing relationships, with a common middle element, into one.

In what follows, we shall examine the relations that satisfy the property

$$aRb \quad \text{and} \quad bRc \implies aRc$$

Definition

Definition: Transitive Relations

A relation R on a set A is called **transitive** if

$$(a, b) \in R \quad \text{and} \quad (b, c) \in R \implies (a, c) \in R$$

for all $a, b, c \in A$

Thinking About the Definition

Think of the transitivity as

Following two arrows always produce an already existing relation in R .

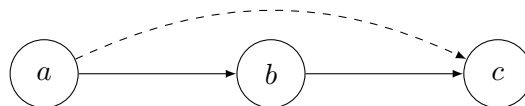
I.e., if travelling in two steps reaches a vertex, there should already be a direct one-step arrow.

A Graphical Interpretation

Imagine travelling along the arrows of a directed graph.

If you can travel from a to b , and then from b to c , there should already be a direct arrow from a to c .

This is precisely the idea of transitivity.



The dashed arrow represents the arrow that must exist if the relation is transitive.

Example 63

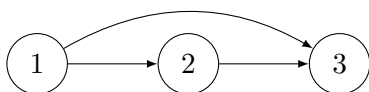
Consider $A = \{1, 2, 3\}$ and

$R = \{(1, 2), (2, 3), (1, 3)\}$.

The only chain of length two is

$$1 \rightarrow 2 \rightarrow 3.$$

Since $(1, 3) \in R$ the relation is transitive.

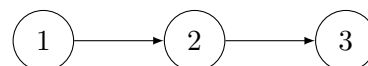


Example 64

Consider $R = \{(1, 2), (2, 3)\}$.

Now $1 \rightarrow 2 \rightarrow 3$ exists, but $(1, 3) \notin R$.

Hence, the relation is not transitive.



Algorithmic Thinking

Algorithm: Testing Transitivity

Given a finite relation R ,

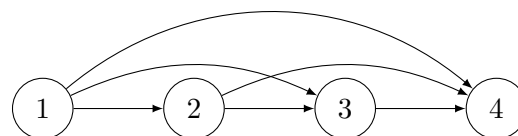
1. List every ordered pair.
2. For every chain $(a, b), (b, c)$ check whether (a, c) belongs to the relation.
3. If every chain satisfies this property, the relation is transitive.
Otherwise, it is not transitive.

Example 65 :Worked Application of the Algorithm

Let $A = \{1, 2, 3, 4\}$ and consider the relation $R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$.

We test every possible chain of length two.

- $(1, 2), (2, 3) \in R \implies (1, 3) \in R$.
- $(1, 2), (2, 4) \in R \implies (1, 4) \in R$.
- $(1, 3), (3, 4) \in R \implies (1, 4) \in R$.
- $(2, 3), (3, 4) \in R \implies (2, 4) \in R$.



Every chain has the required shortcut. Therefore, R is transitive.

Example 66 : A relation that is not transitive

Let $S = \{(1, 2), (2, 3), (3, 4), (1, 3)\}$.

The pairs $(2, 3), (3, 4) \in S$.

from the chain $2 \longrightarrow 3 \longrightarrow 4$, transitivity would require $(2, 4)$ to be in S .

However, $(2, 4) \notin S$. Therefore, S is not transitive.

How to Disprove Transitivity

To prove that a relation is not transitive, it is enough to find one triple a, b, c such that

$$(a, b) \in R, \quad (b, c) \in R,$$

but

$$(a, c) \notin R.$$

Example 67: Loops and Transitivity

Loops do not add extra conditions for transitivity.

Suppose $(a, a) \in R$ and $(a, b) \in R$. The chain $a \longrightarrow a \longrightarrow b$ requires $(a, b) \in R$.

However, this pair is already present.

Similarly, if $(a, b) \in R$ and $(b, b) \in R$ then the chain $a \longrightarrow b \longrightarrow b$ again requires only the already existing pair (a, b) .

Thus, loops do not automatically cause a relation to fail transitivity.

Example 68: Two Opposite Arrows

Suppose $(a, b) \in R$ and $(b, a) \in R$. Then there are two chains:

$$a \longrightarrow b \longrightarrow a \quad \text{and} \quad b \longrightarrow a \longrightarrow b.$$

If R is transitive, these chains require $(a, a) \in R$ and $(b, b) \in R$.

Important Observation

If a transitive relation contains both

$$(a, b) \quad \text{and} \quad (b, a),$$

then it must also contain the loops

$$(a, a) \quad \text{and} \quad (b, b).$$

Example 69: Involving Opposite Arrows

Consider $R = \{(1, 2), (2, 1)\}$ on $A = \{1, 2\}$.

Since $(1, 2), (2, 1) \in R$, transitivity would require $(1, 1) \in R$.

Similarly, it would require $(2, 2) \in R$.

Neither loop is present. Hence, R is not transitive.

However, the relation $S = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ is transitive.

Matrix Interpretation

Let R be a relation on the finite set $A = \{a_1, a_2, \dots, a_n\}$ and let $M_R = (m_{ij})$ be its relation matrix. The entry $m_{ij} = 1$ means $(a_i, a_j) \in R$.

A path of length two from a_i to a_k exists whenever there is some a_j such that $m_{ij} = 1$ and $m_{jk} = 1$.

If R is transitive, then the existence of such a two-step path requires $m_{ik} = 1$

Now, recall that a two-step path in R would mean an edge in $R \circ R$. In particular, as per the matrix interpretation of composition of relations, this would mean that whenever the i, k entry of $M_R \odot M_R$ is 1, the corresponding entry in M_R should also be 1.

Theorem

Theorem: Matrix Interpretation of Transitivity

A relation is transitive if every nonzero entry produced by Boolean squaring of its relation matrix already corresponds to a 1 in the original matrix.

Symbolically,

$$M_R \odot M_R \leq M_R,$$

where $\odot$ denotes Boolean matrix multiplication.

Here the inequality is entrywise. This can be interpreted in any of the following equivalent ways.

1. (i, j) entry of $M_R \odot M_R$ is less than or equal the (i, j) entry of M_R .
2. If (i, j) entry of $M_R \odot M_R$ is 1, then the (i, j) entry of M_R is also 1.

Boolean Matrix Multiplication

Under Boolean matrix multiplication, multiplication is logical “and”, and addition is replaced by logical “or.”

Thus, the (i, k) -entry of $M_R \odot M_R$ is 1 exactly when there is an index j such that $(a_i, a_j) \in R$ and $(a_j, a_k) \in R$.

Example 70: Matrix Interpretation

Consider $M_R = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. The relation is $R = \{(1, 2), (1, 3), (2, 3)\}$.

The Boolean square is $M_R \odot M_R = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

The only two-step path is $1 \longrightarrow 2 \longrightarrow 3$.

Since the original matrix already has a 1 in position $(1, 3)$,

$$M_R \odot M_R \leq M_R.$$

Therefore, R is transitive.

Example 71: Transitive Relations from Mathematics

The following relations often considered in mathematics are **transitive**.

Relation	Reason/Justification
Equality ($=$)	$a = b$ and $b = c$ imply $a = c$
Less Than ($<$)	$a < b$ and $b < c$ imply $a < c$
Less Than or Equal ($\leq$)	$a \leq b$ and $b \leq c$ imply $a \leq c$
Divisibility ($ $)	If $a \mid b$ and $b \mid c$, then there exist integers m, n such that $b = am$ and $c = bn$. Substituting $b = am$, we obtain $c = (am)n = a(mn)$. Since mn is an integer, $a \mid c$. Therefore, divisibility is transitive.
Subset Inclusion	Suppose $A \subseteq B$ and $B \subseteq C$ Then, every element of A belongs to B, and every element of B belongs to C. Hence every element of A belongs to C. Therefore, $A \subseteq C$.
Congruence Modulo n	Suppose $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. Then $n \mid (a - b)$ and $n \mid (b - c)$. Since $a - c = (a - b) + (b - c)$, we have $n \mid (a - c)$. Therefore, $a \equiv c \pmod{n}$.

Example 72: Relations that are usually not transitive

Relation	Reason/Justification
Friendship	Friendship is generally not transitive. Alice may be a friend of Bob, and Bob may be a friend of Carol, without Alice being a friend of Carol.
Parenthood	The relation “is a parent of” is not transitive. If a is a parent of b, and b is a parent of c, then a is a grandparent of c, not a parent of c.

Relation	Reason/Justification
Difference by One	Define a relation on the integers by $aRb \iff a - b = 1$. We have $1R2$ and $2R3$, but $1R3$ is false. Hence the relation is not transitive.

Example 73 : Vacuously Transitive Relations

Consider $R = \{(1, 2)\}$ on $A = \{1, 2, 3\}$.

There is no pair beginning with 2, so there is no chain $(a, b), (b, c) \in R$.

Therefore, no counterexample to transitivity exists.

Hence R is transitive.

Vacuous Truth

A relation may be transitive even if it contains no chains of length two.
The transitivity condition says that every such chain must have a shortcut.
If there are no chains, the condition is automatically satisfied.

Common Mistakes

Common Mistakes

1. Checking only one chain and concluding that the relation is transitive.
Every chain of length two must be checked.
2. Thinking that a relation must contain a chain in order to be transitive.
A relation with no two-step chains is vacuously transitive.
3. Requiring the converse implication.
Transitivity does not say that $(a, c) \in R$ implies the existence of b with $(a, b), (b, c) \in R$.
4. Assuming that a long path alone is sufficient.
The definition is checked through all two-step chains. Repeated use then produces shortcuts for longer paths.
5. Ignoring opposite arrows.
Loops may form chains but they do not add extra conditions.
However, opposite arrows force loops.

Theorem: Longer Chains Have Shortcuts

Theorem

Let R be a transitive relation. If

$$a_0Ra_1, \quad a_1Ra_2, \quad \dots, \quad a_{n-1}Ra_n,$$

then a_0Ra_n .

Proof. We prove the statement by induction on n .

For $n = 1$, the statement is immediate.

Assume that every chain of length n has the required shortcut.

Consider a chain of length $n + 1$:

$$a_0Ra_1, \quad a_1Ra_2, \quad \dots, \quad a_nRa_{n+1}.$$

By the inductive hypothesis, a_0Ra_n .

Since a_nRa_{n+1} transitivity gives a_0Ra_{n+1} .

Thus, every chain of length $n + 1$ also has the required shortcut.

Therefore, by Induction, the result holds for all positive integers n . □

Theorem: Intersection Preserves Transitivity

Theorem

If R and S are transitive relations on A , then $R \cap S$ is transitive.

Proof.

Suppose $(a, b) \in R \cap S$ and $(b, c) \in R \cap S$.

Then $(a, b), (b, c) \in R$ and $(a, b), (b, c) \in S$

Since R is transitive, $(a, c) \in R$.

Since S is transitive, $(a, c) \in S$.

Therefore, $(a, c) \in R \cap S$.

Hence $R \cap S$ is transitive. □

Example 74: Does Union Preserve Transitivity?

The union of two transitive relations need not be transitive.

Let $R = \{(1, 2)\}$ and $S = \{(2, 3)\}$.

Each relation is transitive because neither contains a two-step chain.

However, $R \cup S = \{(1, 2), (2, 3)\}$ is not transitive, since $(1, 3) \notin R \cup S$.

Activity 33: Test Transitivity

Determine whether each relation is transitive. Justify your answer.

1. $R_1 = \{(1, 2), (2, 3), (1, 3)\}$ on $\{1, 2, 3\}$.
2. $R_2 = \{(1, 2), (2, 1)\}$ on $\{1, 2\}$.
3. $R_3 = \{(1, 1), (1, 2), (2, 2)\}$ on $\{1, 2\}$.
4. $R_4 = \{(1, 2), (3, 4)\}$ on $\{1, 2, 3, 4\}$.
5. $R_5 = \{(1, 1), (2, 2), (3, 3)\}$ on $\{1, 2, 3\}$.
6. $R_6 = \{(1, 2), (2, 3), (3, 4), (1, 3), (2, 4), (1, 4)\}$

Activity 34: Find a Counterexample

For each non-transitive relation, identify a triple (a, b, c) satisfying $(a, b), (b, c) \in R$ but $(a, c) \notin R$.

1. $R = \{(1, 2), (2, 4), (3, 1)\}$
2. $S = \{(a, b), (b, c), (c, d), (a, c)\}$
3. The relation “has the same first letter as” on a collection of words.
4. The relation “is the parent of” on a set of people.

Activity 35: Complete the Relation

Add the smallest possible number of ordered pairs to make each relation transitive.

1. $R = \{(1, 2), (2, 3)\}$
2. $R = \{(1, 2), (2, 3), (3, 4)\}$
3. $R = \{(1, 2), (2, 1)\}$
4. $R = \{(a, b), (b, c), (c, d), (a, c)\}$

See “transitive closure” of a relation for a follow-up on these exercises.

Activity 36: Classification Activity

Classify each of the following relations as transitive or not.

1. Equality on $\mathbb{R}$
2. Less than on $\mathbb{Z}$
3. Less than or equal to on $\mathbb{R}$
4. Divisibility on $\mathbb{N}$
5. Congruence modulo 5
6. Friendship
7. Parenthood
8. Subset inclusion

Exercises (ctd.)

29. Prove the following relations are transitive:

- (a) equality
- (b) divisibility
- (c) congruence modulo n
- (d) subset relation
- (e) inverse of a transitive relation
- (f) intersection of transitive relations

30. Give an example showing that the union of two transitive relations need not be transitive.
31. Let R be a symmetric and transitive relation. Prove that if $(a, b) \in R$ then $(a, a) \in R$ and $(b, b) \in R$.
32. Determine whether the difference of two transitive relations must be transitive. Prove your answer or give a counterexample.
33. Determine all transitive relations on the set $A = \{1, 2\}$ and classify them.
34. Let R be transitive. Prove that $R^n \subseteq R$ for every integer $n \geq 1$, where R^n denotes the n -fold composition of R with itself.
35. Let $R = \{(a, b) \in \mathbb{Z}^2 : a - b \text{ is even}\}$. Prove that R is transitive.
36. Let $R = \{(a, b) \in \mathbb{R}^2 : a \leq b + 1\}$. Determine whether R is transitive.
37. Let $R = \{(A, B) : A \cap B \neq \emptyset\}$ on the collection of nonempty subsets sets of $\mathbb{Z}$. Determine whether R is transitive.
38. (Diagnostic Questions) Answer without lengthy calculation.
 - (a) Is the empty relation be transitive?
 - (b) Can a relation with one ordered pair be transitive?
 - (c) Must every transitive relation be reflexive?
 - (d) Must every reflexive relation be transitive?
 - (e) Can a transitive relation contain opposite arrows?
 - (f) If it contains opposite arrows, what additional pairs may be required?
 - (g) Is the union of transitive relations always transitive?
 - (h) Is the intersection of transitive relations always transitive?

Section Summary

Summary: Transitive Relations

- A relation R on A is transitive if $(a, b) \in R$ and $(b, c) \in R$ always imply $(a, c) \in R$.
- Graphically: $a \longrightarrow b \longrightarrow c$ requires the shortcut $a \longrightarrow c$.
- To test transitivity:
 1. Find every chain of length two.
 2. Check whether its shortcut is present.
 3. One missing shortcut disproves transitivity.
- Important facts:
 - A relation with no two-step chains is vacuously transitive.
 - Opposite arrows in a transitive relation force appropriate loops.
 - Intersections of transitive relations are transitive.
 - Unions of transitive relations need not be transitive.

Looking Ahead

A relation may fail transitivity only because certain shortcut pairs are missing.

The **transitive closure** of a relation is obtained by adding the necessary pairs while keeping the relation as small as possible.

We shall study this construction together with reflexive and symmetric closures next.

13 Closures of Relations

A relation may fail to have a particular property because some ordered pairs are missing.

Instead of replacing the relation completely, we may add only the necessary ordered pairs.

This leads to the idea of the **closure** of a relation.

General Idea of a Closure

Let R be a relation on a set A .

A closure of R with respect to a property is the smallest relation containing R that has that property.

In this section, we study:

reflexive closure, symmetric closure, transitive closure.

Reflexive Closure

Suppose R is a relation on A .

To make R reflexive, we must make sure that $(a, a) \in R$ for every $a \in A$.

Thus, we add every missing diagonal pair.

Recall that the identity relation on A is $I_A = \{(a, a) : a \in A\}$.

Thus, we have the following.

Definition: Reflexive Closure

The reflexive closure of R is $R_{\text{ref}} = R \cup I_A$.

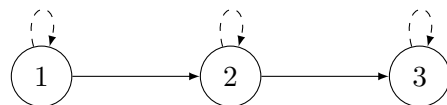
Example 75

Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 3)\}$.

$$R_{\text{ref}} = \{(1, 2), (2, 3), (1, 1), (2, 2), (3, 3)\}$$

The missing diagonal pairs are $(1, 1)$, $(2, 2)$, $(3, 3)$.

Therefore, the reflexive closure is R_{ref} :



The dashed loops are the ordered pairs added to obtain the reflexive closure.

Why Is It the Smallest?

Any reflexive relation on A must contain every pair (a, a) .

Therefore, any reflexive relation containing R must also contain $R \cup I_A$.

Hence $R \cup I_A$ is the smallest reflexive relation containing R .

Algorithmic Thinking

Algorithm: Constructing the Reflexive Closure

Given a relation R on a finite set A :

1. List every element $a \in A$.
2. Check whether $(a, a) \in R$.
3. Add every missing diagonal pair.

Equivalently, $R_{\text{ref}} = R \cup I_A$.

Symmetric Closure

Suppose R contains (a, b) .

For the relation to be symmetric, it must also contain (b, a) .

Thus, we add the reverse of every ordered pair that does not already have its reverse.

Definition: Symmetric Closure

The symmetric closure of R is

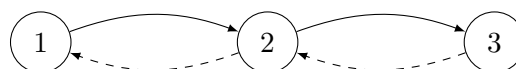
$$R_{\text{sym}} = R \cup R^{-1}.$$

Example 76

Let $R = \{(1, 2), (2, 3)\}$.

Then $R^{-1} = \{(2, 1), (3, 2)\}$.

Therefore, $R_{\text{sym}} = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$.



The dashed arrows are the reverse pairs added to obtain symmetry.

Why Is It the Smallest?

Any symmetric relation containing R must contain the reverse of every pair in R .

Thus, it must contain R^{-1} .

Therefore, it must contain $R \cup R^{-1}$.

Hence $R \cup R^{-1}$ is the smallest symmetric relation containing R .

Algorithmic Thinking

Algorithm: Constructing the Symmetric Closure

Given a relation R :

1. Take each pair $(a, b) \in R$.
2. Check whether $(b, a) \in R$.
3. If not, add (b, a) .

Equivalently, $R_{\text{sym}} = R \cup R^{-1}$

Transitive Closure

Suppose R contains (a, b) and (b, c) .

For R to be transitive, it must also contain (a, c) .

The transitive closure is obtained by adding all missing shortcut pairs.

Note that in the above case $(a, c) \in R \circ R$.

Definition: Transitive Closure

The transitive closure of R , denoted R^+ , is the smallest transitive relation containing R .

Example 77

Let $R = \{(1, 2), (2, 3)\}$.

Since $1 \rightarrow 2 \rightarrow 3$, transitivity requires $(1, 3)$.

Therefore, $R^+ = \{(1, 2), (2, 3), (1, 3)\}$.

Example 78

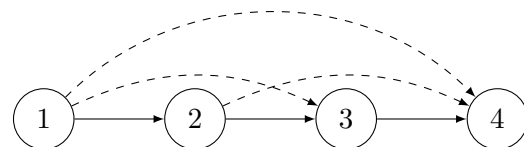
Let $R = \{(1, 2), (2, 3), (3, 4)\}$.

The two-step chains require $(1, 3)$ and $(2, 4)$.

After adding these, the chain $1 \rightarrow 3 \rightarrow 4$ requires $(1, 4)$.

Thus,

$R^+ = \{(1, 2), (2, 3), (3, 4), (1, 3), (2, 4), (1, 4)\}$.



The dashed arrows are the shortcut pairs added to obtain transitivity.

Path Interpretation

The transitive closure relates two elements whenever there is a directed path from one to the other.

Thus, $(a, b) \in R^+$ if there is a path of positive length from a to b in the directed graph of R .

Relation Powers

Define

$$R^1 = R$$

and

$$R^{n+1} = R^n \circ R.$$

Then R^n contains pairs connected by a path of length n .

Therefore, $R^+ = R \cup R^2 \cup R^3 \cup \dots$.

For a finite set, this process eventually stops adding new ordered pairs.

Algorithmic Thinking

Algorithm: Constructing the Transitive Closure

Given a finite relation R :

1. Find every chain $(a, b), (b, c) \in R$.
2. Add (a, c) if it is missing.
3. Repeat the process using the enlarged relation.
4. Stop when no new ordered pairs are produced.

The final relation is the transitive closure R^+ .

Warshall's Algorithm

Warshall's algorithm computes the transitive closure using the relation matrix.

Algorithmic Thinking: Warshall's Algorithm

Suppose $A = \{a_1, \dots, a_n\}$.

Start with

$$W^{(0)} = M_R.$$

For $k = 1, 2, \dots, n$,
define

$$w_{ij}^{(k)} = w_{ij}^{(k-1)} \vee \left(w_{ik}^{(k-1)} \wedge w_{kj}^{(k-1)} \right).$$

Here:

- $\vee$ means logical “or”;
- $\wedge$ means logical “and”.

After the final stage, $W^{(n)}$ is the matrix of the transitive closure.

Warshall Algorithm in Words: At stage k , ask:

Can we travel from a_i to a_j through the intermediate vertex a_k ?

If $a_i \longrightarrow a_k$ and $a_k \longrightarrow a_j$, then record that $a_i \longrightarrow a_j$ is reachable.

Example 79

$$\text{Let } M_R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

This represents $R = \{(1, 2), (2, 3)\}$.

Using vertex 2 as an intermediate vertex, we obtain a path $1 \longrightarrow 2 \longrightarrow 3$.

Therefore, the entry in position $(1, 3)$ becomes 1.

$$\text{The matrix of the transitive closure is } M_{R^+} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Example 80: Comparing the Three Closures

Let $A = \{1, 2, 3\}$ and $R = \{(1, 2), (2, 3)\}$.

Then:

Reflexive Closure :

$$R_{\text{ref}} = \{(1, 2), (2, 3), (1, 1), (2, 2), (3, 3)\}.$$

Symmetric Closure :

$$R_{\text{sym}} = \{(1, 2), (2, 1), (2, 3), (3, 2)\}.$$

Transitive Closure :

$$R^+ = \{(1, 2), (2, 3), (1, 3)\}.$$

Closure	Pairs added
Reflexive closure	Missing loops (a, a)
Symmetric closure	Missing reverse pairs (b, a)
Transitive closure	Missing shortcut pairs (a, c)

Reflexive–Transitive Closure

In some applications, we want a relation to be both reflexive and transitive.

The reflexive–transitive closure of R , often denoted R^* , is

$$R^* = I_A \cup R^+.$$

It represents reachability by a path of length zero or more.

A path of length zero keeps an element at itself, which accounts for the identity relation.

Common Mistakes

Common Mistakes

1. Adding unnecessary ordered pairs.
A closure must be the smallest relation with the required property.
2. Confusing reflexive closure with symmetric closure.
Reflexive closure adds loops. Symmetric closure adds reverse arrows.
3. Stopping the transitive-closure process too early.
New shortcut pairs may create additional chains requiring further shortcuts.
4. Adding loops automatically when constructing the transitive closure.
Loops are added only when transitivity requires them.
5. Assuming the transitive closure is always reflexive. It need not be reflexive.

Exercises (Ctd.)

38. (Diagnostic Questions)

- (a) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (3, 4)\}$. Find the reflexive closure.
- (b) For the same relation, find the symmetric closure.
- (c) Let $R = \{(1, 2), (2, 3)\}$. Find the transitive closure.
- (d) Let $R = \{(1, 2), (2, 3), (3, 4)\}$. Find the transitive closure.
- (e) Let $R = \{(1, 2), (2, 1)\}$. Find the transitive closure.
- (f) Let $R = \{(a, b), (b, c), (c, d)\}$. Find all three closures.

39. (Add the Missing Pairs) For each relation, add the smallest number of ordered pairs needed to obtain the stated closure.

- (a) Make $R = \{(1, 2), (2, 3), (3, 4)\}$ reflexive on $\{1, 2, 3, 4\}$.
- (b) Make $R = \{(1, 2), (2, 3), (3, 4), (4, 4)\}$ symmetric.
- (c) Make $R = \{(1, 2), (2, 3), (1, 4)\}$ transitive.
- (d) Make $R = \{(1, 2), (2, 1)\}$ transitive, symmetric and reflexive.
- (e) Make $R = \{(1, 2), (2, 3), (3, 1)\}$ transitive.

40. (Matrix Exercises)

- (a) The matrix of a relation is $M_R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Find the matrices of:

- i. the reflexive closure;
- ii. the symmetric closure;
- iii. the transitive closure.

(b) Let

$$M_R = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Find the matrix of the transitive closure.

(c) Use Warshall's algorithm to find the transitive closure of

$$M_R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

41. (Proof Exercises) Prove the following.

- (a) $R \cup I_A$ is reflexive.
- (b) Every reflexive relation containing R also contains $R \cup I_A$.
- (c) $R \cup R^{-1}$ is symmetric.
- (d) Every symmetric relation containing R also contains $R \cup R^{-1}$.
- (e) Intersection of all transitive relations containing R is the transitive closure of R .
- (f) $R^+ = \bigcup_{n \geq 1} R^n$.
- (g) $R^* = I_A \cup R^+$ is reflexive and transitive.

42. (Challenges)

- (a) Let $R = \{(1, 2), (2, 3), (3, 1)\}$. Find the transitive closure. Explain why loops appear.
- (b) Let $R = \{(1, 2), (2, 3), (3, 4), (4, 2)\}$. Find the transitive closure.
- (c) Give an example where the symmetric closure is not transitive.
- (d) Give an example where the transitive closure is not symmetric.
- (e) Determine whether taking the symmetric closure and then the transitive closure always gives the same result as taking the transitive closure and then the symmetric closure.
- (f) Let R be already reflexive. What is its reflexive closure?
- (g) Let R be already symmetric. What is its symmetric closure?
- (h) Let R be already transitive. What is its transitive closure?

Section Summary

Summary: Closures of Relations

A closure is the smallest relation containing R that has a desired property.

$$R_{\text{ref}} = R \cup I_A$$

$$R_{\text{sym}} = R \cup R^{-1}$$

$$R^+ = \bigcup_{n \geq 1} R^n$$

The reflexive closure adds missing loops.

The symmetric closure adds missing reverse pairs.

The transitive closure adds missing shortcut pairs and represents reachability by directed paths.

Warshall's algorithm provides a systematic matrix method for computing the transitive closure of a finite relation.

Looking Ahead

We have now studied the main properties and closures of relations.

The next task is to bring everything together to classify mathematical relations.

We shall recognize two fundamental classes of relations:

Reflexive + Symmetric + Transitive $\implies$ Equivalence Relation,
Reflexive + Antisymmetric + Transitive $\implies$ Partial Order.

14 Two Important Classes of Relations

We have studied several properties that a relation may possess:

reflexive, irreflexive, symmetric, antisymmetric, transitive.

Two particular combinations of these properties occur throughout mathematics.

Two Fundamental Structures

A relation that is

reflexive, symmetric and transitive

is called an **equivalence relation**.

A relation that is

reflexive, antisymmetric and transitive

is called a **partial order**.

- Equivalence relations classify objects into groups of objects that are regarded as equivalent (equal).
- Partial orders organize objects according to a notion of precedence, containment, divisibility or size.

Equivalence Relations

Motivation

In many situations, objects may be different but are treated as equivalent because they share an important property.

For example:

- integers may have the same remainder modulo 5;
- students may have the same date of birth;
- fractions may represent the same rational number;
- geometric figures may have the same shape;
- words may have the same length.

Such relationships usually satisfy three properties:

1. every object is equivalent to itself;
2. if one object is equivalent to another, then the reverse is also true;
3. if one object is equivalent to a second, and the second to a third, then the first is equivalent to the third.

Definition

Definition: Equivalence Relation

A relation R on a set A is an **equivalence relation** if it is

1. reflexive:

$$aRa \quad \text{for every } a \in A;$$

2. symmetric:

$$aRb \implies bRa;$$

3. transitive:

$$aRb \text{ and } bRc \implies aRc.$$

Algorithmic Thinking

Algorithm: Testing for an Equivalence Relation

Given a relation R on A :

1. Check that every element has a loop.
2. Check that every arrow has a reverse arrow.
3. Check that every chain of two arrows has the required shortcut.

If all three conditions hold, then R is an equivalence relation.

Example 81

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 4), (4, 3)\}$.
We test the three properties.

Reflexivity: All diagonal pairs $(1, 1), (2, 2), (3, 3), (4, 4)$ belong to R .

Hence R is reflexive.

Symmetry : The pair $(1, 2)$ occurs together with $(2, 1)$, and
the pair $(3, 4)$ occurs together with $(4, 3)$.

The loops are their own reverses.

Hence, R is symmetric.

Transitivity: The elements 1 and 2 are related only to one another, and
the elements 3 and 4 are related only to one another.

All required shortcuts are present.

Therefore, R is transitive.

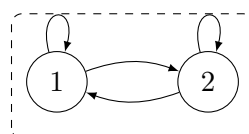
Hence R is an equivalence relation.

Graphical Interpretation

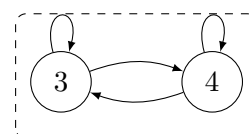
An equivalence relation separates the set into groups. Within each group, every element is related to every other element.

Example 81

The equivalence classes of the above example can be represented as follows.



One equivalence class



Another equivalence class

Example 82: Equality

Equality is an equivalence relation.

- **Reflexivity:** $a = a$
- **Symmetry:** If $a = b$, then $b = a$
- **Transitivity:** $a = b$ and $b = c$, then $a = c$

Example 83: Congruence Modulo n

Define a relation on $\mathbb{Z}$ by

$$aRb \iff a \equiv b \pmod{n}.$$

We show that this is an equivalence relation.

Reflexivity: Since $a - a = 0$ is divisible by n .

Thus, $a \equiv a \pmod{n}$.

Symmetry: If $a \equiv b \pmod{n}$ then $n \mid (a - b)$

Therefore, $n \mid -(a - b) = b - a$.

Thus, $b \equiv a \pmod{n}$.

Transitivity: If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $n \mid (a - b)$ and $n \mid (b - c)$.

Since $a - c = (a - b) + (b - c)$, we obtain $n \mid (a - c)$.

Therefore, $a \equiv c \pmod{n}$.

Hence congruence modulo n is an equivalence relation.

Example 84: A Non-Example

The relation $\leq$ on $\mathbb{R}$ is not an equivalence relation.

It is reflexive and transitive, but it is not symmetric.

For example, $2 \leq 5$ but $5 \not\leq 2$.

Equivalence Classes

Definition: Equivalence Class

Let R be an equivalence relation on A .

For $a \in A$, the equivalence class of a is

$$[a] = \{x \in A : xRa\}.$$

It is the set of all elements that are related to a .

Some authors use $[a]_R$ when the relation must be specified explicitly.

Example 85

For the relation $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 4), (4, 3)\}$, which is an equivalence relation, we have

$$[1] = \{1, 2\}, \quad [2] = \{1, 2\},$$

and

$$[3] = \{3, 4\}, \quad [4] = \{3, 4\}.$$

Notice that different representatives may determine the same equivalence class.

Example 86: Congruence Classes Modulo 3

Congruence modulo 3 divides the integers into three equivalence classes.

These equivalence classes are usually referred to as "congruence classes".

They are:

$$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\},$$

$$[1] = \{\dots, -5, -2, 1, 4, 7, \dots\},$$

$$[2] = \{\dots, -4, -1, 2, 5, 8, \dots\}.$$

These are also written as

$$3\mathbb{Z}, \quad 3\mathbb{Z} + 1, \quad 3\mathbb{Z} + 2.$$

Important Properties of Equivalence Classes

Properties of Equivalence Classes

Let R be an equivalence relation on A .

Then:

1. every equivalence class is nonempty;
2. every element belongs to its own class;
3. if aRb , then

$$[a] = [b];$$

4. if a is not related to b , then

$$[a] \cap [b] = \emptyset;$$

5. two equivalence classes are either equal or disjoint.

Theorem

Theorem

Let R be an equivalence relation on A . If $[a] \cap [b] \neq \emptyset$, then $[a] = [b]$.

Proof. Suppose $[a] \cap [b] \neq \emptyset$.

We will prove that $[a] = [b]$.

As $[a] \cap [b] \neq \emptyset$, let $x \in [a] \cap [b]$. So, xRa and xRb .

Since R is symmetric, we have that aRx and xRb .

Now, by transitivity, aRb , and also bRa (by symmetry).

To prove $[a] = [b]$, we shall prove $[a] \subseteq [b]$ and $[b] \subseteq [a]$.

Suppose $y \in [a]$.

Then yRa and aRb .

By transitivity, yRb

Thus, $y \in [b]$.

Therefore, $[a] \subseteq [b]$

Conversely, suppose $y \in [b]$.

Then yRb and bRa .

By transitivity, yRa

Thus, $y \in [a]$.

Therefore, $[b] \subseteq [a]$

Therefore, $[a] = [b]$. □

Partitions

Definition: Partition

A partition of a set A is a collection of nonempty subsets

$$A_1, A_2, \dots$$

such that:

1. the subsets are pairwise disjoint;
2. their union is all of A .

Thus, every element belongs to exactly one part of the partition.

Example 87: A Partition

Let $A = \{1, 2, 3, 4, 5, 6\}$.

Then $\{\{1, 3, 5\}, \{2, 4, 6\}\}$ is a partition of A .

The two parts are nonempty, disjoint and their union is A .

Example 88: Not a Partition

The collection $\{\{1, 2, 3\}, \{3, 4\}, \{5, 6\}\}$ is not a partition because the first two sets overlap at 3.

Fundamental Theorem

Equivalence Relations and Partitions

Every equivalence relation on a set determines a partition of that set into equivalence classes.

Conversely, every partition of a set determines an equivalence relation.

From an Equivalence Relation to a Partition

Let R be an equivalence relation on A .

The collection of distinct equivalence classes

$$\{[a] : a \in A\}$$

forms a partition of A .

- Every class is nonempty because $a \in [a]$.
- Every element belongs to some class (its class) and hence union is all of A , and
- two classes are either equal or disjoint.

From a Partition to an Equivalence Relation

Suppose $\mathcal{P} = \{A_i : i \in I\}$ is a partition of A .

Define a relation R on A by

$$aRb \iff a \text{ and } b \text{ belong to the same part of } \mathcal{P}.$$

Then R is:

- reflexive, because every element belongs to the same part as itself;
- symmetric, because if a and b are in the same part, then b and a are in the same part;
- transitive, because if a, b are in one part and b, c are in one part, then all three belong to the unique part containing b .

Hence R is an equivalence relation.

Quotient Set

The set of all equivalence classes is called the quotient set of A by R .

It is denoted by

$$A/R.$$

Thus,

$$A/R = \{[a] : a \in A\}.$$

Example 89:

For congruence modulo n ,

$$\mathbb{Z}/\equiv_n = \{[0], [1], \dots, [n-1]\}.$$

This is often denoted by

$$\mathbb{Z}_n \quad \text{or} \quad \mathbb{Z}/n\mathbb{Z}.$$

Exercises (Ctd.)

43. Show that the relation

$$aRb \iff a - b \text{ is even}$$

on $\mathbb{Z}$ is an equivalence relation.

Find its equivalence classes.

44. Define a relation on $\mathbb{R}$ by

$$xRy \iff |x| = |y|.$$

Prove that R is an equivalence relation and describe $[3]$.

45. On the set of ordered pairs of integers, define

$$(a, b)R(c, d) \iff a + d = b + c.$$

Investigate whether R is an equivalence relation.

46. Let $A = \{1, 2, 3, 4, 5, 6\}$.

Construct the equivalence relation determined by the partition

$$\{\{1, 4\}, \{2, 3, 6\}, \{5\}\}.$$

47. List all equivalence classes of congruence modulo 4.

48. Prove that

$$aRb \iff a^2 = b^2$$

is an equivalence relation on $\mathbb{R}$.

49. Let $f : A \rightarrow B$ be a function. Define

$$aRb \iff f(a) = f(b).$$

Prove that R is an equivalence relation.

50. Show that the intersection of two equivalence relations on A is an equivalence relation.

51. Is the union of two equivalence relations necessarily an equivalence relation? Give a proof or a counterexample.

52. Prove that any element of an equivalence class may be used as a representative of the same class:

$$b \in [a] \implies [b] = [a].$$

Partial Orders

Motivation

Some relations do not group objects into equivalent classes. Instead, they organize objects according to an ordering.

Examples include:

- numerical order using $\leq$;
- divisibility among positive integers;
- subset inclusion among sets;
- prerequisite relationships among courses;
- dependency relationships among tasks.

In such an ordering, two objects need not always be comparable (thus the terminology *partial* order).

Definition

Definition: Partial Order

A relation R on a set A is called a **partial order** if it is

1. reflexive;
2. antisymmetric;
3. transitive.

A set A together with a partial order R is called a **partially ordered set**, or a **poset**.

Algorithmic Thinking

Algorithm: Testing for a Partial Order

To determine whether R is a partial order:

1. Check that every element is related to itself.
2. Check that no two distinct elements are related in both directions.
3. Check that every two-step chain has the required shortcut.

Example 90: Less Than or Equal To

The relation $\leq$ on $\mathbb{R}$ is a partial order.

- $a \leq a$, so it is reflexive.
- If $a \leq b$ and $b \leq a$, then $a = b$, so it is antisymmetric.
- If $a \leq b$ and $b \leq c$, then $a \leq c$, so it is transitive.

Example 91: Subset Inclusion

The relation $\subseteq$ on $\mathcal{P}(A)$ is a partial order.

- $X \subseteq X$;
- if $X \subseteq Y$ and $Y \subseteq X$, then $X = Y$;
- if $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$.

Example 92: Divisibility

Consider the divisibility relation on $\mathbb{N}$.

$$aRb \iff a \mid b.$$

- Divisibility is reflexive: $a \mid a$
- Divisibility is transitive: $a \mid b$ and $b \mid c$ imply $a \mid c$
- Divisibility on the positive integers is antisymmetric: $a \mid b$ and $b \mid a$ imply $a = b$

Hence, divisibility is a partial order.

Example 93: A Non-Example

Congruence modulo n is not generally a partial order.

It is reflexive and transitive, but it is not antisymmetric.

For example, $1 \equiv 4 \pmod{3}$ and $4 \equiv 1 \pmod{3}$, but $1 \neq 4$.

Comparable and Incomparable Elements

Let $(A, \preceq)$ be a poset.

Two elements $a, b \in A$ are **comparable** if

$$a \preceq b \quad \text{or} \quad b \preceq a.$$

Otherwise, they are **incomparable**.

Example 94

Under divisibility on $\{1, 2, 3, 6\}$ the elements 2 and 6 are comparable because $2 \mid 6$.

However, 2 and 3 are incomparable because $2 \nmid 3$ and $3 \nmid 2$.

Total Orders

Definition: Total Order

A partial order is a **total order** or **linear order** if every pair of elements is comparable.

Example 95

- The usual order $\leq$ on $\mathbb{R}$ is a total order.
- Divisibility is usually only a partial order because not every pair is comparable.

Hasse Diagrams

A Hasse diagram is a simplified picture of a finite partially ordered set.

Instead of drawing every ordered pair, we *omit information* that can be recovered from reflexivity and transitivity.

Constructing a Hasse Diagram

Algorithm: Constructing a Hasse Diagram

Given a finite poset:

1. Draw one vertex for every element.
2. Remove every loop.
3. Remove every edge implied by transitivity.
4. Remove arrowheads.
5. Place larger elements above smaller elements.

Covering Relation

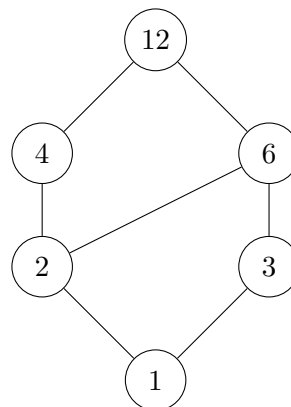
For an partial order $\prec$, an element b **covers** a if $a \prec b$ and there is no element c such that $a \prec c \prec b$.

- Edges in a Hasse diagram represent covering relations.

Example 96: Divisors of 12

Consider the set $D = \{1, 2, 3, 4, 6, 12\}$ ordered by divisibility.

The Hasse diagram is:



The edge from 1 to 2 means $1 \mid 2$.

There is no edge from 1 to 4 because $1 \nmid 4$ and $2 \nmid 4$.

The relation $1 \mid 4$ is already implied by transitivity.

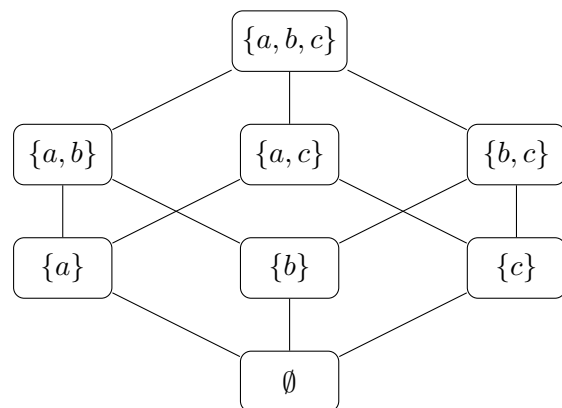
Exampel 97: Power Set Example

Consider the power set of $\{a, b, c\}$ ordered by inclusion.

The power set is:

$$\mathcal{P}(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

The Hasse diagram is:



Minimal, Maximal, Least and Greatest Elements

These concepts must be distinguished carefully.

- **Minimal Element**

An element $m \in A$ is *minimal* if there is no element $x \neq m$ such that $x \preceq m$.

A poset may have several minimal elements.

- **Maximal Element**

An element $m \in A$ is *maximal* if there is no element $x \neq m$ such that $m \preceq x$.

A poset may have several maximal elements.

- **Least Element**

An element $l \in A$ is the *least element* if $l \preceq x$ for every $x \in A$.

A least element, if it exists, is unique.

- **Greatest Element**

An element $g \in A$ is the *greatest element* if $x \preceq g$ for every $x \in A$.

A greatest element, if it exists, is unique.

Common Mistakes

Common Mistake

- A minimal element need not be the least element.
- A maximal element need not be the greatest element.
- A poset may have several minimal or maximal elements, but it can have at most one least element and at most one greatest element.

Example 98

- In the divisibility poset $\{1, 2, 3, 4, 6, 12\}$ the least element is 1, and the greatest element is 12.
- In the poset $\{2, 3, 6\}$ under divisibility, both 2 and 3 are minimal, while 6 is the greatest element.

Upper and Lower Bounds

Let $S \subseteq A$, where $(A, \preceq)$ is a poset.

- **Upper Bound**

An element $u \in A$ is an upper bound of S if $s \preceq u$ for every $s \in S$.

- **Lower Bound**

An element $l \in A$ is a lower bound of S if $l \preceq s$ for every $s \in S$.

- **Least Upper Bound**

A least upper bound of S , denoted $\text{lub}(S)$ or $\sup S$, is an upper bound that is less than or equal to every other upper bound.

- **Greatest Lower Bound**

A greatest lower bound of S , denoted $\text{glb}(S)$ or $\inf S$, is a lower bound that is greater than or equal to every other lower bound.

Example 99

In the divisibility poset of positive integers $(\mathbb{N})$, consider $S = \{4, 6\}$.

A common upper bound is a common multiple.

The least upper bound is $\text{lcm}(4, 6) = 12$.

A common lower bound is a common divisor.

The greatest lower bound is $\text{gcd}(4, 6) = 2$.

Thus, under divisibility,

$$\text{lub}(4, 6) = \text{lcm}(4, 6),$$

and

$$\text{glb}(4, 6) = \text{gcd}(4, 6).$$

A Brief Introduction to Lattices

Definition: Lattice

A poset is called a **lattice** if every pair of elements has

- a least upper bound;
- a greatest lower bound.

The least upper bound of a and b is often written

$$a \vee b,$$

and the greatest lower bound is written

$$a \wedge b.$$

Example 100

- The power set $\mathcal{P}(A)$, ordered by inclusion, is a lattice:

$$X \vee Y = X \cup Y, \qquad X \wedge Y = X \cap Y.$$

- Positive integers ordered by divisibility form a lattice:

$$a \vee b = \text{lcm}(a, b), \qquad a \wedge b = \text{gcd}(a, b).$$

Equivalence Relations and Partial Orders Compared

Feature	Equivalence Relation	Partial Order
Required properties	Reflexive, symmetric, transitive	Reflexive, antisymmetric, transitive
Concept	Groups “equivalent” objects/ Partition	Organizes objects by order
Main structure	Equivalence classes	Hasse diagram
Typical example	Congruence modulo n	Divisibility, $\leq$
Associated collection	Partition	Poset
Two-way relation	Allowed between distinct elements	Forces equality

Exercises (Ctd.)

53. Determine whether each relation is a partial order:

- | | |
|--------------------------------|--|
| (a) equality on $\mathbb{R}$; | (d) divisibility on $\mathbb{N}$; |
| (b) $<$ on $\mathbb{R}$; | (e) congruence modulo 5; |
| (c) $\leq$ on $\mathbb{R}$; | (f) subset inclusion on $\mathcal{P}(A)$. |

54. Draw the Hasse diagram for divisibility on

$$\{1, 2, \dots, 9, 10\}.$$

55. Draw the Hasse diagram for divisibility on the set of divisors of 24

Is this partial order also a total order?

56. Draw the Hasse diagram of $\mathcal{P}(\{a, b, c, d\})$ under inclusion.

57. For each Hasse diagram above, identify:

- minimal elements;
- maximal elements;
- least element (if exist);
- greatest element (if exist).

58. Consider the divisibility poset $A = \{1, 2, 3, 4, 6, 12\}$.

- (a) Find all upper bounds of $\{2, 3\}$.
- (b) Find all lower bounds of $\{4, 6\}$.
- (c) Find the least upper bound of $\{2, 3\}$.
- (d) Find the greatest lower bound of $\{4, 6\}$.
- (e) Does $\{3, 4\}$ have a least upper bound in A ?
- (f) Does every pair of elements in A have a greatest lower bound?

59. (Proof Exercises) Prove the following.

- (a) Prove that $\leq$ is a partial order on $\mathbb{R}$.
- (b) Prove that inclusion is a partial order on $\mathcal{P}(A)$.
- (c) Prove that divisibility is a partial order on the positive integers.
- (d) Prove that a least element, if it exists, is unique.
- (e) Prove that a greatest element, if it exists, is unique.
- (f) Prove that a least upper bound, if it exists, is unique.
- (g) Prove that every least element is minimal.
- (h) Give an example showing that a minimal element need not be least.
- (i) Prove that the intersection of two partial orders on A is a partial order.

60. (Comprehensive Review Questions) Give brief answers:

- (a) Define an equivalence relation.
- (b) Define a partial order.
- (c) Explain the difference between symmetry and antisymmetry.
- (d) Explain how an equivalence relation produces a partition.
- (e) Explain how a partition produces an equivalence relation.
- (f) What is an equivalence class?
- (g) What is a quotient set?
- (h) What information is removed when constructing a Hasse diagram?
- (i) Explain the difference between maximal and greatest.
- (j) Explain the difference between an upper bound and a least upper bound.
- (k) Give an example of a total order.
- (l) Give an example of a partial order that is not total.

Section Summary

Summary

An equivalence relation is

reflexive + symmetric + transitive.

It partitions a set into disjoint equivalence classes.

A partial order is

reflexive + antisymmetric + transitive.

It organizes a set into a poset, which may be represented by a Hasse diagram.

Equivalence relations answer the question

Which objects should be regarded as equivalent?

Partial orders answer the question

How can the objects be organized or compared?

Appendix A: Review Exercises

1. (Diagnostic Questions) Answer each question using the definitions rather than intuition.
 - (a) Must every element of the source set belong to the actual domain?
 - (b) Must every element of the codomain belong to the range?
 - (c) Can one element have several images under a relation?
 - (d) Can an element have no image under a relation?
 - (e) If $R \subseteq A \times B$, which way does R^{-1} go?
 - (f) If $R \subseteq A \times B$, is $\text{Dom}(R^{-1}) = \text{Ran}(R)$?
 - (g) Does $S \circ R = R \circ S$ always hold?
 - (h) What is the matrix of the inverse relation?
 - (i) What is the identity relation on A ?
 - (j) How many relations exist from a finite set A to a finite set B ?
2. (Recall) Answer the following without consulting the notes.
 - (a) Define a relation from A to B .
 - (b) Let $R = \{(1, a), (1, b), (3, c)\}$.
Find $\text{Dom}(R)$ and $\text{Ran}(R)$.
 - (c) Explain the difference between codomain and range.
 - (d) Find the inverse of $R = \{(1, 2), (2, 3), (3, 1)\}$.
 - (e) Let $R = \{(1, a), (2, b)\}$ and $S = \{(a, x), (b, y)\}$.
Find $S \circ R$.
 - (f) Write the identity relation on $A = \{1, 2, 3\}$.
 - (g) State one important identity involving inverse relations.
 - (h) In $S \circ R$, which relation is applied first?
3. (Basic Concepts) Let $A = \{1, 2\}$ and $B = \{a, b, c\}$.
 - (a) Write down $A \times B$.
 - (b) How many relations are possible from A to B ?
 - (c) Determine whether each statement is true or false:
 - i. $(1, a) \in A \times B$
 - ii. $(a, 1) \in A \times B$
 - iii. $A \times B = B \times A$
 - (d) Let $R = \{(1, a), (1, c), (2, b), (2, c)\}$. Find
 - i. $\text{Dom}(R)$
 - ii. $\text{Ran}(R)$
 - iii. $R(1)$
 - iv. $R^{-1}(c)$
4. (Representing Relations) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (1, 4), (2, 3), (3, 3), (4, 1)\}$.
 - (a) Construct the arrow diagram.
 - (b) Construct the relation table.
 - (c) Construct the relation matrix.
 - (d) Draw the directed graph.
 - (e) Identify every loop in the graph.

5. (Domain, Codomain and Range) Let $A = \{1, 2, 3, 4, 5\}$, $B = \{a, b, c, d, e\}$ and $R = \{(1, a), (1, c), (2, b), (4, c), (5, a)\}$.

Find:

- | | |
|------------------------|------------------------|
| (a) the source set, | (e) $R(1)$, |
| (b) the codomain, | (f) $R^{-1}(3)$, |
| (c) the actual domain, | (g) $R(\{1, 2, 4\})$. |
| (d) the range, | |

6. (Inverse Relations) Let $R = \{(1, a), (2, b), (3, a), (3, c)\}$

- | | |
|--|--|
| (a) Find R^{-1} . | (d) Construct M_R and $M_{R^{-1}}$. |
| (b) Find $(R^{-1})^{-1}$. | (e) Verify $M_{R^{-1}} = M_R^T$ |
| (c) Verify that $\text{Dom}(R^{-1}) = \text{Ran}(R)$ | |

7. (Composition) Let $R = \{(1, a), (1, b), (2, c), (3, a)\}$ and $S = \{(a, x), (b, y), (c, x), (c, z)\}$. Find:

- | | |
|--|---------------------------------|
| (a) $S \circ R$, | (c) the range of $S \circ R$, |
| (b) the actual domain of $S \circ R$, | (d) the matrix of $S \circ R$. |

8. (Identity Relations) Let $A = \{1, 2, 3, 4\}$.

- | | |
|--|--|
| (a) Write I_A . | (d) Let $R = \{(1, 2), (2, 3), (4, 1)\}$. |
| (b) Construct the matrix M_{I_A} . | Verify directly that $R \circ I_A = R$ and |
| (c) Draw the directed graph of I_A . | $I_A \circ R = R$. |

9. (Proof Exercises) Prove the following statements.

- For every relation R , $(R^{-1})^{-1} = R$.
- If $R \subseteq A \times B$, then $\text{Dom}(R^{-1}) = \text{Ran}(R)$.
- If $R \subseteq A \times B$, then $\text{Ran}(R^{-1}) = \text{Dom}(R)$.
- For relations R and S , $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$.
- For relations R and S , $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.
- For relations R and S , $(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}$.
- For composable relations R and S , $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.
- If $R \subseteq A \times B$, prove $R \circ I_A = R$.
- If $R \subseteq A \times B$, prove $I_B \circ R = R$.
- Prove associativity of relation composition: $T \circ (S \circ R) = (T \circ S) \circ R$
Here, $R : A \rightarrow B$, $S : B \rightarrow C$ and $T : C \rightarrow D$.

10. Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ and $C = \{x, y\}$.

Define $R = \{(1, a), (1, b), (2, c), (3, a)\}$ and $S = \{(a, x), (b, y), (c, x), (c, y)\}$.

- Find the actual Domain and Range of R .
- Find $R(1)$, $R(2)$ and $R^{-1}(3)$.

- (c) (Inverse Relation) Find R^{-1} , $\text{Dom}(R^{-1})$ and $\text{Ran}(R)$,
- (d) (Composition) Find $S \circ R$.

11. Some challenges:

- (a) Construct relations R and S such that $S \circ R \neq R \circ S$.
Be careful to ensure both compositions are defined.
- (b) Find a nonempty relation R satisfying $R^{-1} = R$.
Describe the graphical meaning of this condition.
- (c) Let $R = \{(1, 2), (2, 3), (3, 4)\}$.
Compute $R \circ R$, then compute $R \circ R \circ R$.
Interpret the results using paths in the directed graph.
- (d) For a relation R on A , define $R^2 = R \circ R$.
What does $(a, b) \in R^2$ mean graphically?
- (e) Define recursively

$$R^1 = R \quad \text{and} \quad R^{n+1} = R \circ R^n.$$

Interpret, $(a, b) \in R^n$ using directed paths.

- (f) Let $R = \{(x, y) \in \mathbb{Z}^2 : y = x + 1\}$.
Find R^{-1} and $R \circ R$.
Describe each relation using a rule.
- (g) Let $R = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$.
Find R^{-1} .
Is R^{-1} a function? Explain.
- (h) Suppose $|A| = m$ and $|B| = n$.
Prove that the number of relations from A to B is 2^{mn} .

12. (Capstone Activity: Analyze a Relation Completely)

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 3), (2, 4), (3, 2), (4, 1), (4, 4)\}$.

Complete a full analysis of R .

- | | |
|--|---|
| (a) State the Cartesian product in which R is contained. | (h) Find $R(\{1, 4\})$ and $R^{-1}(\{3, 4\})$. |
| (b) Write R as a relation table. | (i) Find R^{-1} . |
| (c) Construct the relation matrix. | (j) Construct the matrix of R^{-1} . |
| (d) Draw an arrow diagram. | (k) Verify $M_{R^{-1}} = M_R^T$. |
| (e) Draw the directed graph. | (l) Compute $R \circ R$. |
| (f) Find $\text{Dom}(R)$ and $\text{Ran}(R)$. | (m) Compare $R \circ R$ with R . |
| (g) Find $R(1)$, $R(2)$, $R^{-1}(3)$ and $R^{-1}(4)$. | (n) Verify $R \circ I_A = R$. |

13. (Independent Investigation)

Do an independent investigation on the following topics.

- (a) Relations in relational databases.
- (b) Directed graphs as representations of relations.
- (c) Social networks as mathematical relations.

- (d) Family relationships and inverse relations.
- (e) Road networks and composition of relations.
- (f) Relations in recommendation systems.
- (g) Boolean matrices and computer algorithms.
- (h) Relation composition in database queries.
- (i) Relations and state-transition systems.
- (j) Relations as the foundation of functions.

Appendix B: Mathematical Toolbox

Important Definitions

- **Cartesian Product**

For sets A and B , $A \times B = \{(a, b) : a \in A, b \in B\}$.

If A and B are finite, then $|A \times B| = |A||B|$.

- **Binary Relation**

A relation from A to B is any subset $R \subseteq A \times B$.

If $A = B$, then R is called a relation on A .

- **Relation Defined by a Predicate**

A relation may be written in the form $R = \{(a, b) \in A \times B : P(a, b)\}$ where $P(a, b)$ is a statement involving a and b .

- **Actual Domain**

$\text{Dom}(R) = \{a \in A : \exists b \in B, (a, b) \in R\}$

It consists of the first coordinates that actually occur in the relation.

- **Codomain**

If $R \subseteq A \times B$, then B is the codomain or target set.

The codomain must be specified as part of the definition of the relation and need not equal the range..

- **Range**

$\text{Ran}(R) = \{b \in B : \exists a \in A, (a, b) \in R\}$

The range is the set of codomain elements actually reached by the relation.

Therefore, $\text{Ran}(R) \subseteq B$. Equality need not hold.

- **Image of an Element**

For $a \in A$, $R(a) = \{b \in B : (a, b) \in R\}$.

For a general relation, $R(a)$ may contain no elements, one element or several elements.

- **Image of a Subset**

For $S \subseteq A$, $R(S) = \{b \in B : \exists a \in S, (a, b) \in R\}$.

Equivalently, $R(S) = \bigcup_{a \in S} R(a)$

- **Relation Matrix**

Let $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_n\}$. Let $R \subseteq A \times B$. The matrix of R is $M_R = (m_{ij})$, where

$$m_{ij} = \begin{cases} 1, & (a_i, b_j) \in R, \\ 0, & (a_i, b_j) \notin R. \end{cases}$$

- **Directed Graph of a Relation**

For a relation R on A , the directed graph has one vertex for each element of A .

There is an arrow from a to b precisely when $(a, b) \in R$.

- **Inverse Relation**

If $R \subseteq A \times B$, then $R^{-1} = \{(b, a) : (a, b) \in R\}$.

Thus, $(a, b) \in R \iff (b, a) \in R^{-1}$.

For finite relations, $M_{R^{-1}} = M_R^T$.

- **Composition of Relations**

Let $R \subseteq A \times B$ and $S \subseteq B \times C$.

Then $S \circ R = \{(a, c) \in A \times C : \exists b \in B, (a, b) \in R \text{ and } (b, c) \in S\}$

In the expression $S \circ R$, the relation R is used first.

- **Identity Relation**

The identity relation on A is $I_A = \{(a, a) : a \in A\}$.

It relates every element of A to itself and to no other element.

It satisfies $R \circ I_A = R$ and $I_B \circ R = R$, whenever $R \subseteq A \times B$.

- **Power of a Relation**

For a relation R on A ,

$$R^1 = R \quad \text{and} \quad R^{n+1} = R^n \circ R.$$

The pair (a, b) belongs to R^n when there is a path of length n from a to b .

- **Reflexive Relation**

A relation R on A is reflexive if $(a, a) \in R$ for every $a \in A$.

Equivalently, $I_A \subseteq R$.

- **Irreflexive Relation**

A relation R on A is irreflexive if $(a, a) \notin R$ for every $a \in A$.

A relation that is not reflexive need not be irreflexive.

- **Symmetric Relation**

A relation R on A is symmetric if $(a, b) \in R \implies (b, a) \in R$ for all $a, b \in A$.

Equivalently, $R = R^{-1}$.

- **Antisymmetric Relation**

A relation R on A is antisymmetric if $(a, b) \in R$ and $(b, a) \in R \implies a = b$.

Thus, two distinct elements cannot be related in both directions.

- **Transitive Relation**

A relation R on A is transitive if $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$ for all $a, b, c \in A$.

Graphically, every path of length two must have the corresponding shortcut arrow.

- **Reflexive Closure**

The reflexive closure of a relation R on A is $R_{\text{ref}} = R \cup I_A$.

It is the smallest reflexive relation containing R .

- **Symmetric Closure**

The symmetric closure of R is $R_{\text{sym}} = R \cup R^{-1}$.

It is the smallest symmetric relation containing R .

- **Transitive Closure**

The transitive closure of R , denoted R^+ , is the smallest transitive relation containing R .

It may be written as $R^+ = \bigcup_{n \geq 1} R^n$.

Thus, $(a, b) \in R^+$ if and only if there is a directed path of positive length from a to b .

- **Reflexive–Transitive Closure**

The reflexive–transitive closure of R is $R^* = I_A \cup R^+$.

It represents reachability by a path of length zero or more.

- **Equivalence Relation**

A relation R on A is an equivalence relation if it is *reflexive*, *symmetric*, and *transitive*.

Equivalence relations group elements that are regarded as “equivalent”.

- **Equivalence Class**

Let R be an equivalence relation on A .

For $a \in A$, the equivalence class of a is $[a] = \{x \in A : xRa\}$.

It is the set of all elements equivalent to a .

- **Quotient Set**

The quotient set of A by an equivalence relation R is the set of all distinct equivalence classes:

$$A/R = \{[a] : a \in A\}.$$

- **Partition**

A partition of a set A is a collection $\mathcal{P}$ of nonempty subsets of A such that:

1. distinct members of $\mathcal{P}$ are disjoint;
2. the union of all members of $\mathcal{P}$ is A .

Equivalently, every element of A belongs to exactly one part.

- **Equivalence Relation–Partition Correspondence**

Every equivalence relation on A determines a partition of A into equivalence classes.

Conversely, every partition of A determines an equivalence relation by declaring two elements equivalent when they belong to the same part.

- **Partial Order**

A relation R on A is a partial order if it is *reflexive*, *antisymmetric*, and *transitive*.

- **Partially Ordered Set**

A set A , together with a partial order $\preceq$, is called a partially ordered set or **poset**.

It is written as $(A, \preceq)$.

- **Comparable Elements**

Two elements a, b of a poset are comparable if $a \preceq b$ or $b \preceq a$.

Otherwise, they are incomparable.

- **Total Order**

A partial order is a total order or linear order if every pair of elements is comparable.

- **Covering Relation**

In a poset, an element b covers a if $a \prec b$ and there is no element c such that $a \prec c \prec b$.

- **Hasse Diagram**

A Hasse diagram is a simplified representation of a finite poset.

Loops, arrowheads, and edges implied by transitivity are omitted.

If $a \prec b$, then b is drawn above a .

- **Minimal Element**

An element $m \in A$ is minimal if there is no $x \in A$, with $x \neq m$, such that $x \preceq m$.

- **Maximal Element**

An element $m \in A$ is maximal if there is no $x \in A$, with $x \neq m$, such that $m \preceq x$.

- **Least Element**

An element $l \in A$ is the least element if $l \preceq x$ for every $x \in A$.

A least element, if it exists, is unique.

- **Greatest Element**

An element $g \in A$ is the greatest element if $x \preceq g$ for every $x \in A$.

A greatest element, if it exists, is unique.

- **Upper Bound**

Let $S \subseteq A$, where $(A, \preceq)$ is a poset.

An element $u \in A$ is an upper bound of S if $s \preceq u$ for every $s \in S$.

- **Lower Bound**

An element $l \in A$ is a lower bound of S if $l \preceq s$ for every $s \in S$.

- **Least Upper Bound**

An element u is the least upper bound of S if:

1. u is an upper bound of S ;
2. $u \preceq v$ for every upper bound v of S .

It is denoted by $\text{lub}(S)$ or $\sup S$.

- **Greatest Lower Bound**

An element l is the greatest lower bound of S if:

1. l is a lower bound of S ;
2. $v \preceq l$ for every lower bound v of S .

It is denoted by $\text{glb}(S)$ or $\inf S$.

- **Lattice**

A poset is a lattice if every pair of elements has both:

1. a least upper bound;
2. a greatest lower bound.

For two elements a and b , these are often denoted by $a \vee b$ and $a \wedge b$ respectively.

Important Notations

Notation	Meaning
Basics of Relations	
$A \times B$	Cartesian product of A and B .
$R \subseteq A \times B$	R is a relation from A to B .
$(a, b) \in R$	The ordered pair (a, b) belongs to R .
aRb	a is related to b .
$\text{Dom}(R)$	Actual domain of R .
$\text{Ran}(R)$	Range of R .
$R(a)$	Image of the element a .
$R(S)$	Image of the subset S .

Notation	Meaning
Operations on Relations	
R^{-1}	Inverse relation.
$S \circ R$	Composition of R followed by S .
I_A	Identity relation on A .
M_R	Matrix representing the relation R .
R^n	The n -th power of the relation R , obtained by repeated composition.
Properties and Closures	
R_{ref}	Reflexive closure of R .
R_{sym}	Symmetric closure of R .
R^+	Transitive closure of R .
R^*	Reflexive-transitive closure of R .
Equivalence Relations	
$[a]$	Equivalence class containing a .
$[a]_R$	Equivalence class of a with respect to R .
A/R	Quotient set (the set of all equivalence classes).
$a \equiv b \pmod{n}$	a is congruent to b modulo n .
$\mathbb{Z}_n$	The set of congruence classes modulo n .
Partial Orders	
$(A, \preceq)$	A partially ordered set (poset).
$a \preceq b$	a precedes (or is less than or equal to) b in a partial order.
$a \prec b$	$a \preceq b$ and $a \neq b$ (strict order).
Bounds and Lattices	
$\text{lub}(S)$	Least upper bound of S .
$\text{glb}(S)$	Greatest lower bound of S .
$\sup S$	Supremum (least upper bound) of S .
$\inf S$	Infimum (greatest lower bound) of S .
$a \vee b$	Join (least upper bound) of a and b .
$a \wedge b$	Meet (greatest lower bound) of a and b .
$\text{gcd}(a, b)$	Greatest common divisor of a and b .
$\text{lcm}(a, b)$	Least common multiple of a and b .

Six Representations of a Relation

A relation may be represented in several equivalent forms.

- | | | |
|----------------------|-------------------|--------------------|
| 1. Predicate or rule | 3. Arrow diagram | 5. Relation matrix |
| 2. Ordered pairs | 4. Relation table | 6. Directed graph |

Changing the representation does not change the relation.
Each representation contains the same relational information.

Important Identities

The following identities summarize many of the results developed throughout this chapter.

Inverse Relations

$$(R^{-1})^{-1} = R.$$

$$(R \cap S)^{-1} = R^{-1} \cap S^{-1}.$$

$$M_{R^{-1}} = M_R^T.$$

$$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}.$$

$$(R \cup S)^{-1} = R^{-1} \cup S^{-1}.$$

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$

Composition of Relations

$$R \circ I_A = R.$$

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

$$I_B \circ R = R.$$

$$R^{m+n} = R^m \circ R^n.$$

Properties of Relations

$$R \text{ is symmetric} \iff R = R^{-1}.$$

$$R \text{ is irreflexive} \iff R \cap I_A = \emptyset.$$

$$R \text{ is reflexive} \iff I_A \subseteq R.$$

$$R \text{ is transitive} \iff R \circ R \subseteq R.$$

Closures

$$R_{\text{ref}} = R \cup I_A.$$

$$R_{\text{sym}} = R \cup R^{-1}.$$

$$R^+ = \bigcup_{n \geq 1} R^n.$$

$$R^* = I_A \cup R^+.$$

Equivalence Relations

$$aRb \iff [a] = [b].$$

$$A/R = \{[a] : a \in A\}.$$

$$[a] \cap [b] \neq \emptyset \iff [a] = [b].$$

Bounds in a Divisibility Poset

$$a \vee b = \text{lcm}(a, b).$$

$$a \wedge b = \text{gcd}(a, b).$$

Appendix C: Revision Checklist

Skill	Check
Sets, Cartesian Products and Basic Relations	
Construct a Cartesian product.	☐
Determine the cardinality of a finite Cartesian product.	☐
Define a binary relation as a subset of a Cartesian product.	☐
Determine the number of possible relations from one finite set to another.	☐
Write a relation using ordered pairs.	☐
Write a relation using predicate or set-builder notation.	☐
Distinguish between a relation from A to B and a relation on A .	☐
Representations of Relations	
Construct an arrow diagram.	☐
Construct a relation table.	☐
Construct a relation matrix.	☐
Draw the directed graph of a relation.	☐
Convert between ordered pairs, tables, matrices, arrow diagrams and digraphs.	☐
Interpret entries of a relation matrix.	☐
Recover a relation from its matrix or directed graph.	☐
Domain, Codomain, Image and Range	
Identify the source set and codomain of a relation.	☐
Find the actual domain of a relation.	☐
Distinguish codomain from range.	☐
Find the range of a relation.	☐
Find the image of an element.	☐
Find the image of a subset.	☐
Determine domain and range from a relation matrix.	☐
Operations on Relations	
Construct an inverse relation.	☐
Interpret the inverse relation using an arrow diagram or digraph.	☐
Use $M_{R^{-1}} = M_R^T$.	☐
Compute the composition of two relations.	☐
Interpret composition using two-step paths.	☐
Compute composition using relation matrices and Boolean multiplication.	☐

Skill	Check
Operations on Relations (Ctd.)	
Construct and use the identity relation.	☐
Compute unions, intersections and differences of relations.	☐
Prove elementary identities involving inverse and composition.	☐
Properties of Relations	
Determine whether a relation is reflexive.	☐
Determine whether a relation is irreflexive.	☐
Distinguish “not reflexive” from “irreflexive.”	☐
Determine whether a relation is symmetric.	☐
Use $R = R^{-1}$ to characterize symmetry.	☐
Determine whether a relation is antisymmetric.	☐
Distinguish symmetric from antisymmetric.	☐
Determine whether a relation is transitive.	☐
Identify a counterexample showing that a relation is not transitive.	☐
Interpret transitivity using paths of length two.	☐
Test transitivity using a relation matrix.	☐
Classify a relation according to all its properties.	☐
Closures of Relations	
Construct the reflexive closure of a relation.	☐
Use $R_{\text{ref}} = R \cup I_A$.	☐
Construct the symmetric closure of a relation.	☐
Use $R_{\text{sym}} = R \cup R^{-1}$.	☐
Construct the transitive closure by adding missing shortcut pairs.	☐
Interpret the transitive closure as reachability.	☐
Use relation powers $R, R^2, R^3, \dots$ to describe the transitive closure.	☐
Apply Warshall’s algorithm to a small relation matrix.	☐
Explain why a closure is the smallest enlargement having the required property.	☐
Equivalence Relations and Partitions	
Define an equivalence relation.	☐
Test whether a relation is an equivalence relation.	☐
Prove that congruence modulo n is an equivalence relation.	☐
Find the equivalence class of an element.	☐
Determine all distinct equivalence classes.	☐

Skill	Check
Equivalence Relations and Partitions (Ctd.)	
Prove that two equivalence classes are either equal or disjoint.	☐
Construct a partition from an equivalence relation.	☐
Construct an equivalence relation from a partition.	☐
Explain the equivalence-relation–partition correspondence.	☐
Construct and interpret a quotient set.	☐
Partial Orders and Hasse Diagrams	
Define a partial order and a partially ordered set.	☐
Test whether a relation is a partial order.	☐
Distinguish a partial order from a total order.	☐
Determine whether two elements are comparable.	☐
Construct a Hasse diagram from a finite partial order.	☐
Identify covering relations.	☐
Find minimal and maximal elements.	☐
Distinguish minimal from least.	☐
Distinguish maximal from greatest.	☐
Find upper and lower bounds of a subset.	☐
Find a least upper bound and a greatest lower bound.	☐
Interpret gcd and lcm in a divisibility poset.	☐
Determine whether a finite poset is a lattice.	☐
Proof and Problem-Solving Skills	
Prove equality of relations using mutual inclusion.	☐
Use a counterexample to disprove a relational property.	☐
Prove that the intersection of transitive relations is transitive.	☐
Prove that a least or greatest element is unique.	☐
Explain the role of reflexivity, symmetry, antisymmetry and transitivity in major classes of relations.	☐

Appendix D: Concept Map

